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A unified neural framework for long-term time series forecasting and granger-style causal analysis

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ABSTRACT

Reliable long-term time series forecasting (LTSF) in complex multivariate systems requires both effective long-range sequence modeling and informative analysis of cross-variable dependencies. Existing methods usually optimize these goals separately, leading to a mismatch between predictive performance and causal interpretability. To address this issue, we propose Mamba with Kolmogorov-Arnold Network and Jacobian Regularization (MKJR), a unified framework for mutual reinforcement between LTSF and Granger-style causal analysis. MKJR combines a Kolmogorov-Arnold Network (KAN)-based encoder-decoder with an asymmetric bidirectional dual-stream Mamba backbone. The KAN modules perform variable-wise nonlinear embedding and forecasting readout, while exposing encoder-side source-lag structural priors and decoder-side target-aware readout preferences. Built on this representation layer, the dual-stream Mamba captures fine-grained local temporal signatures and broader cross-variable dependency patterns within the same predictor. During training, MKJR uses a matrix-free random-projection Jacobian regularizer to constrain local sensitivity and encourage smoother, more target-relevant lagged predictive dependencies. During evaluation, MKJR fuses encoder-side priors, decoder-side readout preference, and Jacobian sensitivity into lag-resolved Granger-style dependence scores for graph recovery. In this way, better predictive representations provide more reliable dependence evidence, while dependence-aware regularization suppresses diffuse sensitivities and improves forecasting robustness. The resulting design is well suited to complex nonlinear multivariate systems with long forecasting horizons and distribution shifts. Experiments on multiple datasets across forecasting and causal analysis settings show that MKJR achieves the best overall performance and remains highly competitive on individual benchmarks, demonstrating practical value in complex industrial settings.

1. Introduction

Long-term time series forecasting (LTSF) is a fundamental problem in many engineering applications, including sequence prediction, anomaly detection, and missing-value imputation. Despite substantial progress in deep learning, it remains difficult to achieve accurate long-range dependency modeling, linear scalability, interpretable nonlinear representation, and causality-aware learning within a unified framework (Wu et al., 2024; Chen et al., 2023).

Existing neural architectures provide only partial solutions. Recurrent models are often limited in modeling very long dependencies, Transformer-based models capture global interactions effectively but suffer from quadratic complexity, Convolutional Neural Network (CNN)-style approaches are efficient yet constrained by fixed receptive patterns, and linear or Multilayer Perceptron (MLP)-based models are scalable but often insufficient for highly nonlinear multivariate temporal dependencies. However, the joint challenge of scalable long-horizon forecasting and structurally meaningful causal interpretation remains insufficiently addressed.

Among recent sequence models, Mamba provides a promising direction by introducing input-adaptive selective state-space modeling, which enables efficient long-range sequence learning with linear complexity (Gu and Dao, 2023). Mamba-based forecasting variants have further shown strong ability in multivariate forecasting and channel interaction modeling (Liang et al., 2024; Ahamed and Cheng, 2024). Nevertheless, Mamba alone does not directly provide interpretable nonlinear transformation or explicit causal-consistency-oriented learning.

A complementary direction is offered by Kolmogorov–Arnold

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Nomenclature

LTSF	Long-term Time Series Forecasting
KANs	Kolmogorov-Arnold Networks
SSMs	State Space Models
VAR	Vector Autoregressive
RNNs	Recurrent Neural Networks
MLP	Multilayer Perceptron
CNN	Convolutional Neural Network
AI	Artificial Intelligence
MKJR	Mamba-KAN with Jacobian Regularization

Networks (KANs), which replace fixed scalar weights with learnable univariate functions and thus provide a more flexible and interpretable nonlinear approximation mechanism than conventional MLPs (Liu et al., 2024b). Recent KAN-based studies further suggest their suitability for structured and nonlinear time-series modeling (Lee et al., 2024a; Han et al., 2024; Huang et al., 2025). These developments indicate that Mamba and KAN are naturally complementary: the former is effective for efficient long-sequence state evolution, whereas the latter is advantageous for expressive and interpretable nonlinear mapping.

Another key issue is causal consistency. Incorporating causal information into forecasting has attracted increasing attention because it can improve both robustness and interpretability. Existing studies span statistical and neural Granger causality, Jacobian-based sensitivity analysis, and graph-based temporal causal modeling, where graph neural networks (GNNs) are used to capture structured cross-variable interactions and relation propagation in multivariatetemporal systems (Runge et al., 2019; Bussmann et al., 2021; Cheng et al., 2024a). However, many of these methods remain standalone tools and are not tightly integrated into high-performance LTSF models.

To address these limitations, we propose a unified framework for long-term forecasting and lag-resolved Granger-style causal analysis. The framework combines a KAN-based encoder-decoder with a heterogeneous bidirectional dual-stream Mamba backbone, so that variable-wise nonlinear structure, local temporal detail, and broader cross-variable dependencies can be modeled within the same forecasting architecture. During training, a matrix-free random-projection Jacobian objective regularizes local sensitivity and encourages more stable and target-relevant lagged predictive dependencies. During evaluation, structural priors from the KAN encoder, decoder-side readout preference, and Jacobian sensitivity evidence are fused to obtain continuous lag-resolved Granger-style dependence scores. In this way, better predictive representations provide more reliable dependence evidence, while dependence-aware regularization improves forecasting robustness and selectivity within a unified forecasting framework.

The main contributions of this work are summarized as follows.

(1) We propose MKJR as a unified framework in which both long-horizon forecasting and lag-resolved Granger-style dependence analysis are built on the same trained multivariate predictor, rather than relying on separate models for prediction and analysis.
(2) We design a forecasting architecture that combines a KAN-based encoder-decoder with an asymmetric bidirectional dual-stream Mamba backbone. In this design, KAN modules preserve variable-wise nonlinear temporal structure and provide structural priors and readout preferences, while the Mamba streams capture local temporal detail and broader cross-variable predictive dependencies.
(3) We introduce a matrix-free random-projection Jacobian regularizer that constrains local sensitivity without explicitly forming a dense Jacobian during training. This regularizer encourages smoother and more target-relevant lagged predictive dependencies, improving both forecasting robustness and the reliability of subsequent Granger-style dependence analysis.
(4) Extensive experiments across standard benchmark datasets and real industrial datasets suggest that MKJR is effective across both LTSF and causal-analysis settings. The results indicate that the proposed framework is not limited to controlled benchmarks and remains promising for real multivariate industrial systems.

The remainder of this paper is organized as follows. Section 2 reviews related work on Artificial Intelligence (AI)-driven predictive modeling, neural forecasting architectures, and neural causality analysis. Section 3 presents the proposed MKJR framework. Section 4 reports the experimental results and discussion. Section 5 concludes the paper and outlines limitations and future directions.

2. Related work

2.1. AI-driven predictive intelligence in complex systems

Recent artificial intelligence research has moved well beyond generic representation learning toward predictive intelligence for complex engineering and dynamical systems. In applied scientific modeling, machine learning and neural-network-based methods have been used for nonlinear fluid prediction, thermal nanofluid analysis, image encryption, fractional-order language-learning models, tuberculosis (TB)-virus dynamical analysis, and hybrid intelligent optimization, showing that modern AI can serve as a flexible tool for approximation, simulation, and prediction in highly nonlinear settings (Abbasi et al., 2025a, 2025b, 2025c, 2025d; Aamir et al., 2025; Raja et al., 2025). In parallel, AI-based predictive learning has also been extended to energy and power applications, including reactive power optimization, renewable-resource power distribution and forecasting, model-based reinforcement learning for electric vehicle (EV) charging, wind power prediction, electricity forecasting, and photovoltaic prediction, further indicating a broader shift toward data-driven, adaptive, and decision-oriented intelligent systems (Abbas et al., 2024; Alnuman et al., 2025; Alshahr et al., 2026; Hanifi et al., 2024; Sun et al., 2024; Wu et al., 2024).

Although these studies confirm the growing relevance of AI in complex prediction tasks, they are largely domain-specific and have not extensively explored the joint study of multivariate long-term time series forecasting (LTSF) and causal analysis.

2.2. Neural architecture trends for long-term time series forecasting

Within time-series research, LTSF architectures have evolved from linear mapping and convolutional interaction models to Transformer-style and multiscale mixing frameworks that aim to balance forecasting accuracy, efficiency, and long-range dependency modeling (Chen et al., 2023; Zeng et al., 2023; Li et al., 2023a; Liu et al., 2022; Zhang and Yan, 2023; Li et al., 2023a; Wang et al., 2024). Another major line is state-space modeling: classical State Space Models (SSMs) provide a principled description of latent state evolution (Hamilton, 1994), while S4 and S4D make long-sequence modeling more practical (Gu et al., 2021, 2022). More recent forecasting-oriented Mamba variants, including S-Mamba (Wang et al., 2025), Bi-Mamba4TS (Liang et al., 2024), FMamba (Ma et al., 2024), and TimeMachine (Ahamed and Cheng, 2024), further explore variable tokenization, bidirectional modeling, fast channel interaction, and channel-mixing strategies for multivariate forecasting.

Recent KAN-based extensions provide another important direction for nonlinear and structured time-series learning. Representative studies include KAN 2.0 (Lee et al., 2024a), DropKAN (Altarabichi, 2024), KAN4TSF (Han et al., 2024), Hippo-KAN (Lee et al., 2024), TimeKAN (Huang et al., 2025), and AdaKAN (Wu et al., 2025), which show that

KAN modules can be combined with structured temporal modeling, frequency decomposition, and adaptive spectral mechanisms for long-term forecasting.

2.3. Neural causality and causality-aware forecasting

A third line of work concerns causality in time series. Classical Granger causality defines a variable as causal if its past improves the prediction of another variable beyond the target's own history (Granger, 1969). This idea has been extended to nonlinear and high-dimensional settings through statistical and neural formulations, including nonlinear causal association analysis, attention-based CNNs, recurrent units, MLP/Recurrent Neural Networks (RNN)-based neural Granger models, additive autoregressive structures, and history-dependent temporal causal learning (Runge et al., 2019; Nauta et al., 2019; Khanna and Tan, 2019; Tank et al., 2021; Bussmann et al., 2021; Gong et al., 2022). In addition, Jacobian regularization and related differential criteria have been used to characterize model sensitivity and support nonlinear causal discovery, including robust Jacobian regularization, Jacobian-based nonlinear ICA discovery, and Jacobian-based Granger analysis (Hoffman et al., 2019; Reizinger et al., 2023; Suryadi et al., 2023).

Recent studies have further broadened this landscape through synthetic benchmarking, irregular time-series causal discovery, orientation-aware causal structure learning, Bayes-factor-based causal evidence, graph-Transformer-based multi-scale causality analysis, flexible time-window Granger discovery, and Jacobian-regularizer-based neural Granger causality (Cheng et al., 2023, 2024a, 2024b; Oravecz and Vandekerckhove, 2024; Wu and Kang, 2024; Zhang et al., 2024a; Zhou et al., 2024). Meanwhile, causality-aware forecasting studies indicate that incorporating causal structure can improve robustness and generalization under distribution shift, as shown in transferable forecasting under causal conditional shift, causality-informed energy forecasting, and causal-perspective time-series modeling (Li et al., 2023b; Sharma et al., 2024; Zhang et al., 2024b). Recent studies have advanced both neural causal analysis and causality-aware forecasting. Yet their tighter integration with long-sequence forecasting and interpretable nonlinear representation remains limited. This motivates the unified design considered in this study.

3. Methodology

This section details the main components of MKJR and describes how they are combined for long-term time series forecasting and post-hoc causal interpretation. An overview of the forecasting architecture is shown in Fig. 1, while the overall MKJR workflow that links training-time Jacobian regularization with evaluation-time lag-resolved causal analysis and graph recovery is summarized in Fig. 3. Our forecasting backbone is a Mamba-based bidirectional dual-stream architecture designed to model long-range temporal dependencies at complementary temporal resolutions. KAN modules are used as the encoder and decoder to transform the input window into latent representations and map the fused hidden states to the forecasting horizon. Residual connections are introduced to facilitate feature fusion across streams and to stabilize optimization. During training, we use a Jacobian-based regularization term to penalize excessive local sensitivity through stochastic matrix-

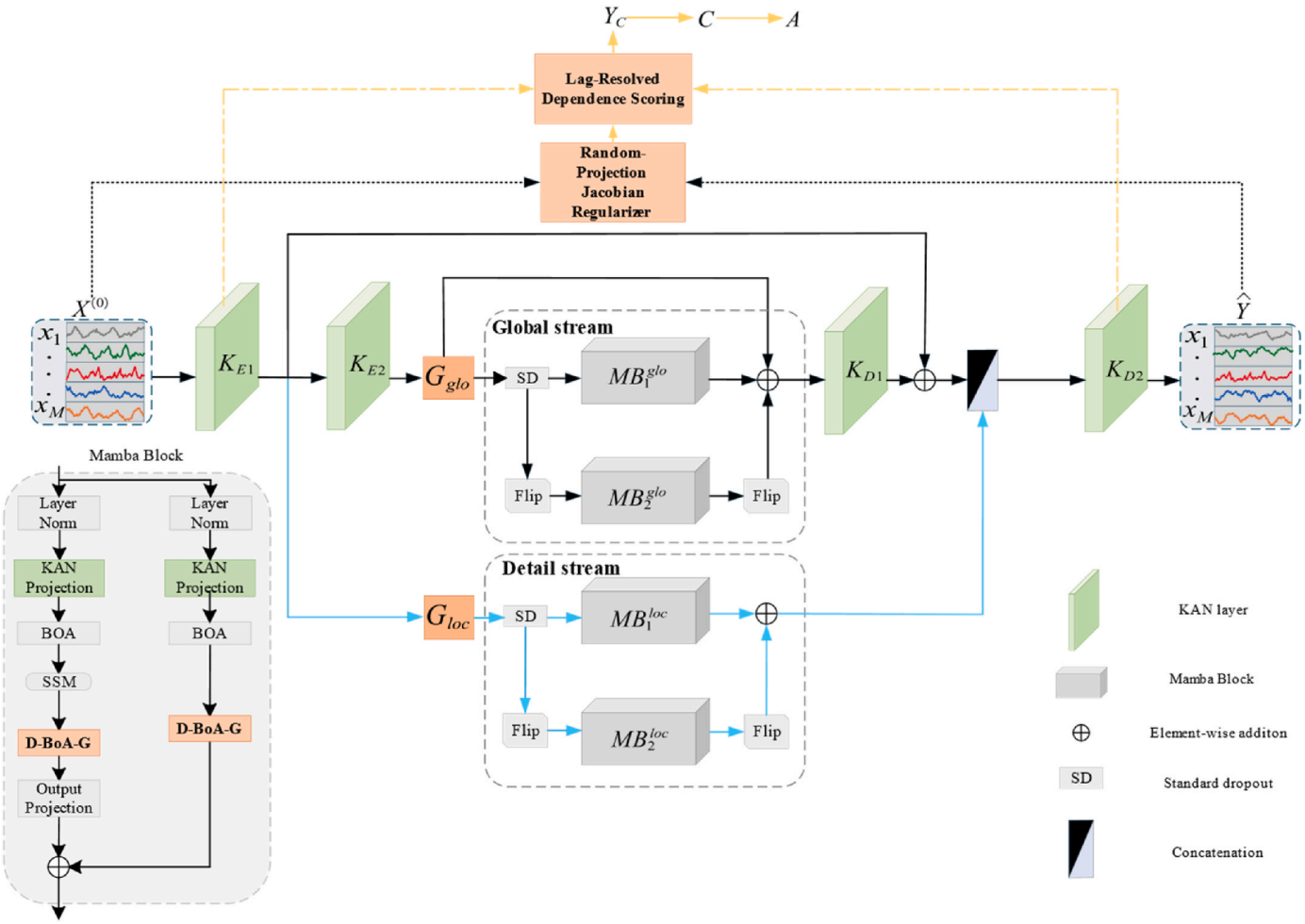


Fig. 1. Overview of the forecasting architecture.

free estimation. After training, we optionally compute a dense Jacobian tensor for post-hoc lagged influence analysis under the fitted predictor. This separation allows the model to support forecasting and interpretability within a unified pipeline, while keeping training-time regularization distinct from evaluation-time influence analysis.

3.1. Data input and preprocessing

We denote the b-th multivariate input window by $X_b \in \mathbb{R}^{M\times L}$, where M is the number of variables and L is the look-back length. A mini-batch is written as $X = [X_b]_{b=1}^{B} \in \mathbb{R}^{B\times M\times L}$. Before entering the network, each input window is normalized by $\mathrm{Norm}(\cdot)$, and the normalized batch input is denoted by $X^{(0)} = \mathrm{Norm}(X) \in \mathbb{R}^{B\times M\times L}$. Accordingly, the normalized b-th sample is written as $X_b^{(0)} \in \mathbb{R}^{M\times L}$. Following common practice in multivariate time-series forecasting, we adopt reversible instance normalization (RevIN) (Kim et al., 2021) as the default preprocessing module. Since RevIN is not a defining architectural component of MKJR, its empirical necessity is examined separately in the ablation study. The primary forecasting objective is to predict the next T steps, whereas the lag-resolved influence scores used for causal analysis are derived only after training and are not treated as a direct supervised output of the model.

3.2. KAN-based representation learning and structural readout

After normalization, the input window is passed through KAN-based embedding layers before entering the Mamba backbone. Compared with standard MLP layers, KAN modules use spline-parameterized learnable edge functions to model nonlinear transformations. In our architecture, they are used to compress variable-wise temporal windows into lower-dimensional latent features and to decode the fused hidden states to the forecasting horizon. These properties motivate the use of KAN modules. In MKJR, KAN modules are used not only for input embedding and output decoding, but also for extracting an encoder-side source-lag prior, producing a decoder-side variable-level readout summary, and serving as the lightweight projection inside the modified Mamba block.

Starting from the normalized batch input $X^{(0)} \in \mathbb{R}^{B\times M\times L}$, the encoder processes each variable-wise trajectory of length L independently. For notational simplicity, we write the KAN mappings in batch form, although each stage is applied independently to each sample window $X_b^{(0)}$. The two encoding stages are written compactly as:

$$Z^{(1)} = K_{E1}\left(X^{(0)}\right), Z^{(2)} = K_{E2}\left(Drop\left(Z^{(1)}\right)\right), \tag{1}$$

where $K_{E1} : \mathbb{R}^{L}\to\mathbb{R}^{N_1}$ and $K_{E2} : \mathbb{R}^{N_1}\to\mathbb{R}^{N_2}$ denote the first and second KAN encoding stages, respectively, with $N_1 > N_2$.

To expose which source–lag groups are structurally retained by the encoder, we summarize the first-stage KAN edge functions associated with source variable i at lag α. Let $w_{h,i,\alpha}^{\mathrm{base}}$ denote the scalar base coefficient and $c_{h,i,\alpha,:}^{\mathrm{spline}}$ denote the spline coefficient vector that connect lag α of variable i to hidden unit h in the first-stage KAN layer. We then directly define the normalized source–lag structural prior as:

$$\pi_{i,\alpha} = \frac{\left(\sum_{h=1}^{N_1}\left[\left(w_{h,i,\alpha}^{\mathrm{base}}\right)^2 + \| c_{h,i,\alpha,:}^{\mathrm{spline}} \|_2^2\right]\right)^{\frac{1}{2}}}{\sum_{\beta=1}^{L}\left(\sum_{h=1}^{N_1}\left[\left(w_{h,i,\beta}^{\mathrm{base}}\right)^2 + \| c_{h,i,\beta,:}^{\mathrm{spline}} \|_2^2\right]\right)^{\frac{1}{2}} + \varepsilon} = \frac{q_{i,\alpha}}{\sum_{\beta=1}^{L} q_{i,\beta} + \varepsilon}, \pi_{i,\alpha} \in [0,1]. \tag{2}$$

Here, $q_{i,\alpha}$ denotes the unnormalized source–lag group magnitude used for training-time structural sparsity, while $\pi_{i,\alpha}$ is its normalized form used later as a structural prior in evaluation-time causal analysis. This definition aggregates all first-stage KAN parameters emitted from lag α of source variable i across the N_1 hidden units into a single group-level strength, and then normalizes it over all lag positions of the same source variable. Thus, $\pi_{i,\alpha}$ reflects the relative retention strength of lag α within variable i. It is not interpreted as a target-specific causal score by itself, but only as a normalized encoder-side structural prior.

After the Mamba backbone, a symmetric two-stage KAN decoder maps the latent hidden states back to the forecasting horizon and is denoted by K_{D1} and K_{D2}. In addition to the encoder-side source-lag structural prior, the first mixing stage of the final decoder K_{D2} also provides a decoder-induced variable-level readout summary. Let $\widetilde{W}^{\mathrm{dec}}$ denote the effective grouped coefficient table on the source-grouped input side of this stage, and let $\mathscr{G}_i$ be the input-channel group associated with source variable i. We define the normalized decoder-side variable-level readout preference as:

$$s_{j,i}^{\mathrm{dec}} = \frac{\sum_{u=1}^{T}\sum_{m\in\mathscr{G}_i}\left|\widetilde{W}_{(j,u),m}^{\mathrm{dec}}\right|}{\sum_{i'=1}^{M}\sum_{u=1}^{T}\sum_{m\in\mathscr{G}_{i'}}\left|\widetilde{W}_{(j,u),m}^{\mathrm{dec}}\right| + \varepsilon}, s_{j,i}^{\mathrm{dec}} \in [0,1]. \tag{3}$$

Unlike $\pi_{i,\alpha}$, which is extracted from the first-stage KAN encoder and serves as a target-agnostic source-lag structural prior, $s_{j,i}^{\mathrm{dec}}$ is obtained post hoc from the final decoder and captures a target-specific yet lag-agnostic variable-level readout preference.

To regularize all KAN-based modules in the proposed architecture, we apply KAN-specific spline dropout within each KAN layer and standard dropout between successive stages. This applies to both the KAN encoder-decoder and the lightweight KAN projection embedded in the modified Mamba block. As shown in Fig. 2, each KAN stage consists of a spline-parameterized nonlinear mapping followed by a projection to the target latent dimension. For readability, these operators are compactly denoted by K_{E1}, K_{E2}, K_{D1}, and K_{D2}, and the corresponding Mamba-side KAN projection is denoted by K_P. Its detailed formulation and motivation are deferred to Section 3.3.

3.3. Asymmetric dual-stream mamba backbone with KAN projection

The Mamba-based heterogeneous bidirectional dual-stream architecture forms the core forecasting backbone of MKJR for LTSF. While the KAN encoder provides nonlinear variable-wise tokenization, the Mamba backbone is used to model predictive dependencies across the resulting variable-token sequence. This design preserves the complementary roles of the KAN encoder and the Mamba backbone: the former summarizes temporal information within each variable, whereas the latter captures predictive interactions across variables.

To make the two streams functionally distinct, we adopt an asymmetric preconditioning design. The higher-dimensional latent tokens $Z^{(1)}$ are assigned to a detail stream, while the more compressed latent tokens $Z^{(2)}$ are assigned to a global stream. The detail stream is modulated by variable-wise local descriptors extracted from the normalized input window so that local temporal signatures are emphasized before compression, whereas the global stream is modulated by compressed latent context so that broader cross-variable interactions are emphasized on a compact representation. The resulting preconditioned tokens are written as:

$$\widetilde{Z}^{loc} = Z^{(1)} \odot \mathrm{Broadcast}_{N_1}\left(\sigma\left(\Phi_{\mathrm{spec}}\left(X^{(0)}\right)\right)\right) = Z^{(1)} \odot G^{loc}, \tag{4}$$

$$\widetilde{Z}^{glo} = Z^{(2)} \odot \sigma\left(\Gamma\left(\left[Z^{(2)}, \mathrm{Pool}_v\left(Z^{(2)}\right)\right]\right)\right) = Z^{(2)} \odot G^{glo}. \tag{5}$$

Here, G^{loc} and G^{glo} denote the local and global gates, respectively. Moreover, $\Phi_{\mathrm{spec}}(\cdot)$ extracts a variable-wise local descriptor from the normalized input window, $\mathrm{Broadcast}_{N_1}(\cdot)$ expands it along the latent dimension to match the N_1-channel detail tokens, $\mathrm{Pool}_v(\cdot)$ denotes variable-wise pooling, and $\Gamma(\cdot)$ is a lightweight fusion mapping.

Given an input token matrix Z, we define Mamba block as:

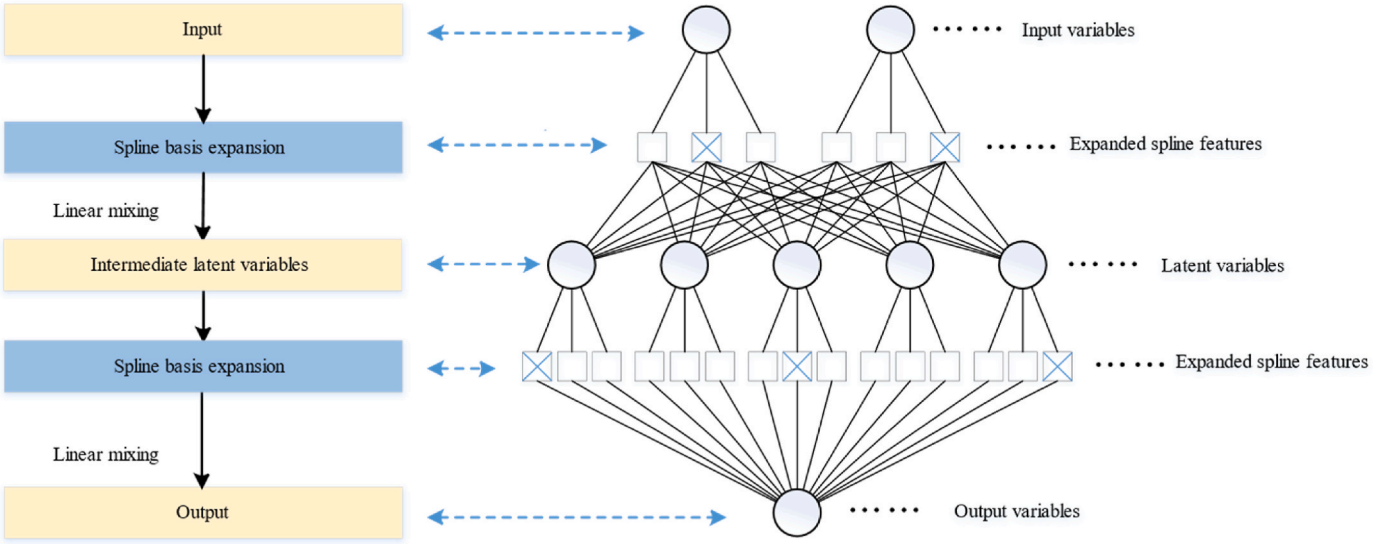


Fig. 2. KAN layer used in MKJR.

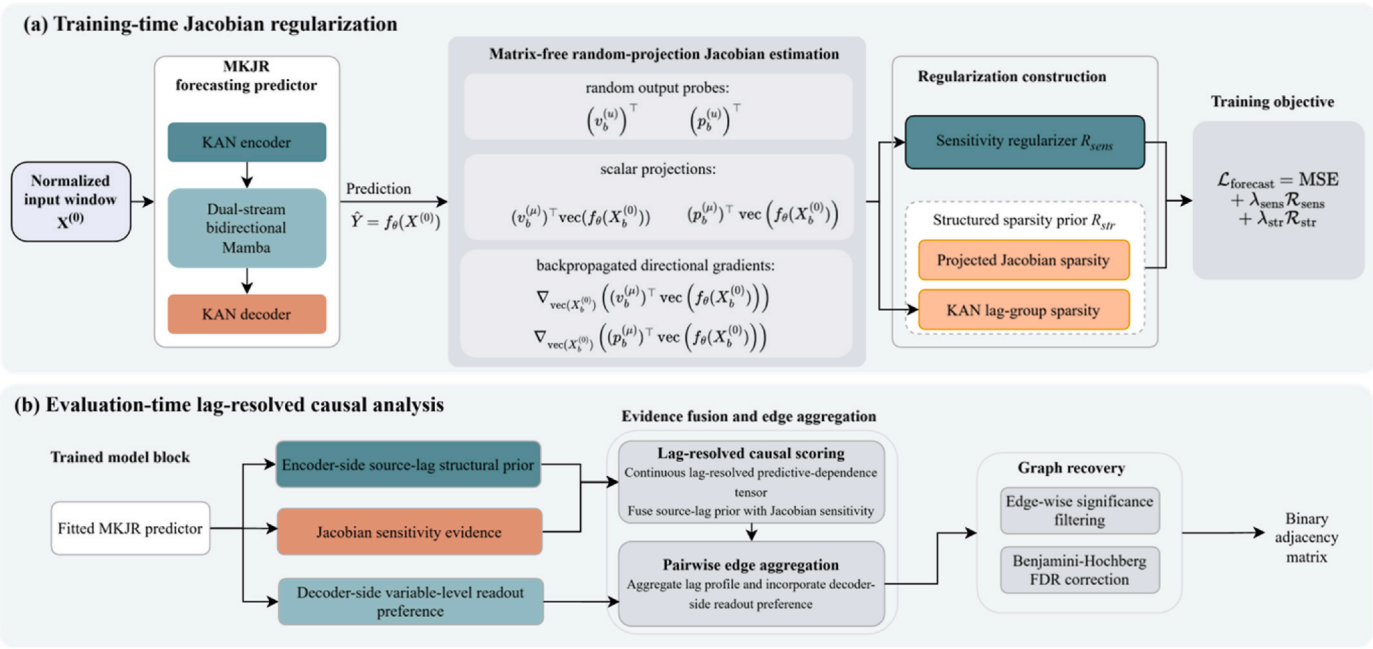


Fig. 3. Overview of the MKJR framework.

$$G_{\mathrm{DB}}(Z) = Z \odot \sigma\left(\sum_{p=1}^{P} \mathrm{Softmax}(\rho(Z))\psi_p(Z)\right), \tag{6}$$

$$MB(Z) = W_o(G_{\mathrm{DB}}(\mathrm{SSM}(\Psi_{\mathrm{BoA}}(K_P(\mathrm{LN}(Z)))))) + G_{\mathrm{DB}}(\Psi_{\mathrm{BoA}}(K_P(\mathrm{LN}(Z)))), \tag{7}$$

Here, $LN(\cdot)$ denotes a pre-normalization applied to latent token representations inside the modified Mamba block. Unlike the input-level RevIN in Section 3.1, which mitigates instance-level distribution shift on raw windows, this LayerNorm mainly stabilizes the subsequent K_P-based nonlinear transformation and optimization. Moreover, K_P denotes the KAN projection, $\Psi_{BoA}(\cdot)$ denotes the Bag-of-Activations (BoA) transformation, $G_{DB}(\cdot)$ denotes the Dynamic BoA Gate (D-BoA-G), and W_o denotes the output projection. Compared with the standard Mamba block, the convolutional pre-filter is removed, since local temporal signatures have already been emphasized by the asymmetric local/global preconditioning before entering the dual-stream backbone. We therefore use a KAN projection followed by BoA to provide more flexible nonlinear adaptation than a fixed activation, and introduce D-BoA-G as a time-series-friendly dynamic recalibration mechanism for both the state-mixed branch and the parallel nonlinear shortcut. In this way, the same modified $MB(\cdot)$ can be shared by both streams, while their functional difference is still determined by the asymmetric preconditioning outside the block.

After asymmetric preconditioning, each stream is processed by a forward Mamba block and a reversed-order Mamba block. The standard low-level selective state-space update is omitted here because it is unchanged by our method and does not constitute the methodological novelty. We therefore focus on the stream-level formulation:

$$H^{(1)} = MB_1^{\mathrm{loc}}\left(\mathrm{Drop}\left(\widetilde{Z}^{loc}\right)\right) + \mathrm{Rev}\left(MB_2^{\mathrm{loc}}\left(\mathrm{Rev}\left(\widetilde{Z}^{loc}\right)\right)\right), \tag{8}$$

$$H^{(2)} = MB_1^{\mathrm{glo}}\left(\mathrm{Drop}\left(\widetilde{Z}^{glo}\right)\right) + \mathrm{Rev}\left(MB_2^{\mathrm{glo}}\left(\mathrm{Rev}\left(\widetilde{Z}^{glo}\right)\right)\right) + Z^{(2)}. \tag{9}$$

Here, MB_1^{loc} and MB_2^{loc} denote the forward and reversed Mamba blocks in the detail stream, respectively, while MB_1^{glo} and MB_2^{glo} denote their counterparts in the global stream. $\mathrm{Rev}(\cdot)$ reverses the variable-token order, and $\mathrm{Drop}(\cdot)$ denotes dropout. The residual design is intentionally asymmetric: the detail stream is kept residual-free so that the high-resolution tokens are more selectively transformed after local preconditioning, whereas the global stream uses an additional residual path to preserve coarse global context and stabilize dependency modeling under stronger compression.

For the b-th sample, the final decoder readout is written as:

$$\widehat{\mathrm{Y}}_b = K_{D2}\left(\mathrm{Concat}\left(H^{(1)}, K_{D1}\left(H^{(2)}\right) + Z^{(1)}\right)\right). \tag{10}$$

Here, K_{D1} first maps the global-stream output back to the N_1-dimensional latent space, and K_{D2} is applied row-wise to the fused token matrix.

3.4. Matrix-free Jacobian Regularization for model training

Directly constructing the full input-output Jacobian at every training iteration is computationally impractical for multivariate long-sequence forecasting. Therefore, instead of explicitly forming the dense Jacobian during optimization, we adopt a matrix-free random-projection strategy and regularize only projected directional responses of the fitted predictor. Let $f_\theta(X_b^{(0)}) = \widehat{Y}_b$ denote the forecasting function parameterized by θ, where $X_b^{(0)} \in \mathbb{R}^{M\times L}$ is the normalized input window and $\widehat{\mathrm{Y}}_b \in \mathbb{R}^{M\times T}$ is the corresponding prediction. We use $\mathrm{vec}(\cdot)$ to stack all entries of a matrix into a column vector.

To regularize the predictor without explicitly forming the dense input-output Jacobian, we adopt a matrix-free random-projection strategy. Specifically, we introduce two complementary projected Jacobian penalties:

$$\mathscr{R}_{\mathrm{sens}} = \frac{1}{B\, n_{\mathrm{proj}}} \sum_{b=1}^{B} \sum_{\mu=1}^{n_{proj}} \left\| \nabla_{\mathrm{vec}\left(X_b^{(0)}\right)} \left(\left(v_b^{(\mu)}\right)^{\top} \mathrm{vec}\left(f_\theta\left(X_b^{(0)}\right)\right) \right) \right\|_2^2, \quad v_b^{(\mu)} \sim \mathcal{N}(0, \mathrm{I}), \tag{11}$$

$$\mathscr{R}_{\mathrm{str}} = \frac{1}{B\, n_{\mathrm{proj}}} \sum_{b=1}^{B} \sum_{\mu=1}^{n_{proj}} \left\| \nabla_{\mathrm{vec}\left(X_b^{(0)}\right)} \left(\left(p_b^{(\mu)}\right)^{\top} \mathrm{vec}\left(f_\theta\left(X_b^{(0)}\right)\right) \right) \right\|_1 + \frac{1}{ML} \sum_{i=1}^{M} \times \sum_{\alpha=1}^{L} q_{i,\alpha}, p_b^{(\mu)} \sim \mathrm{Rademacher}(\pm 1). \tag{12}$$

Here, $v_b^{(\mu)}$ and $p_b^{(\mu)}$ are probe vectors in the output space, $\mathrm{vec}(\cdot)$ stacks all entries of a matrix into a column vector, b indexes samples in the mini-batch, and μ indexes independent probe realizations. The scalar $\left(v_b^{(\mu)}\right)^{\top} \mathrm{vec}\left(f_\theta\left(X_b^{(0)}\right)\right)$ or $\left(p_b^{(\mu)}\right)^{\top} \mathrm{vec}\left(f_\theta\left(X_b^{(0)}\right)\right)$ projects the full multi-output prediction onto a random direction, and the gradient of this scalar with respect to the input approximates one directional slice of the full Jacobian. In this way, Eqs. (11) and (12) define two projected Jacobian penalties that regularize local Jacobian energy and sparsity structure without explicitly forming the dense Jacobian matrix.

For the analysis of Eq. (11), define the sample-wise input-output Jacobian $J_b = \frac{\partial \mathrm{vec}\left(f_\theta\left(X_b^{(0)}\right)\right)}{\partial \mathrm{vec}\left(X_b^{(0)}\right)}$. By the chain rule, $\nabla_{\mathrm{vec}\left(X_b^{(0)}\right)} \left(\left(v_b^{(\mu)}\right)^{\top} \mathrm{vec}\left(f_\theta\left(X_b^{(0)}\right)\right) \right) = J_b^{\top} v_b^{(\mu)}$, and therefore the inner term in Eq. (11) can be written as $\| J_b^{\top} v_b^{(\mu)} \|_2^2 = \left(v_b^{(\mu)}\right)^{\top} J_b J_b^{\top} v_b^{(\mu)}$. Hence, Eq. (11) is a standard Hutchinson-type estimator of $\| J_b \|_F^2$, which is unbiased under isotropic probes. Moreover, when Gaussian probes are used, averaging over n_{proj} independent probes reduces the estimator variance proportionally to $1/n_{\mathrm{proj}}$. In contrast, the projected ℓ_1 term in Eq. (12) is used as a stochastic directional sparsity surrogate rather than as a classical unbiased trace estimator. Therefore, we do not claim the same closed-form variance expression for this term, and instead assess its practical stability through the probe-budget sensitivity analysis in the experiments.

Importantly, the two components of $\mathscr{R}_{\mathrm{str}}$ are not treated as separate objectives but as complementary parts of one structured sparsity prior: projected Jacobian sparsity and KAN lag-group sparsity. They act on different stages of the same predictive pathway and jointly encourage selective dependence patterns. Here, $q_{i,\alpha}$ is the source–lag group magnitude defined in Section 3.2.

The final training objective is defined as the element-wise mean squared error over all batches, target variables, and forecast horizons:

$$\mathscr{L}_{\mathrm{forecast}} = \frac{1}{BMT} \sum_{b=1}^{B} \sum_{j=1}^{M} \sum_{u=1}^{T} \left(\widehat{y}_{b,j,u} - y_{b,j,u} \right)^2 + \lambda_{\mathrm{sens}} \mathscr{R}_{\mathrm{sens}} + \lambda_{\mathrm{str}} \mathscr{R}_{\mathrm{str}}. \tag{13}$$

Here, λ_{sens} and λ_{str} balance the forecasting loss against the two regularization terms. Apart from these two outer coefficients, we do not introduce additional sub-term balancing parameters inside $\mathscr{R}_{\mathrm{str}}$, since both components are already normalized according to their own structural scales. Specifically, the projected Jacobian penalties are averaged over samples and probe realizations, and the explicit structural sparsity term is normalized over source-lag groups. This design keeps the objective tightly coupled to the training process, avoids over-parameterizing the regularization scheme, and reduces additional tuning complexity in the main method. A dedicated sensitivity analysis of λ_{sens} and λ_{str} is provided in the experimental section.

In practice, the probe budget n_{proj} is kept small during training and is increased only when the stochastic estimate becomes unstable. The Jacobian-based objective in this section is used only to regularize the fitted predictor during training. It encourages smooth and selective predictive dependence patterns, but it does not by itself provide formal causal identifiability guarantees.

3.5. Granger-style causal analysis and graph recovery

The Jacobian-based objective in Section 3.4 is used only during training to regularize the fitted predictor. After convergence, we perform evaluation-time lag-resolved causal analysis under the trained model. Let $x_{t+1-\alpha,i}$ denote the α-th lag of input variable i, where $\alpha = 1,\ldots, L$, and let $\widehat{y}_{t+u,j}$ denote the forecast of output variable j at horizon u, where $u = 1,\ldots,T$. We define the final target-specific lag-resolved causal score as:

$$Y^{\mathrm{causal}}_{j\leftarrow i,\alpha} = \pi_{i,\alpha} \frac{1}{T} \sum_{u=1}^{T} \left| \frac{\partial \widehat{y}_{t+u,j}}{\partial x_{t+1-\alpha,i}} \right|, \quad Y^{\mathrm{causal}} \in \mathbb{R}^{M\times M\times L}. \tag{14}$$

Here, $\pi_{i,\alpha}$ is the source-lag structural prior extracted from the first-stage KAN encoder, and the averaged absolute Jacobian term measures the local predictive sensitivity of the fitted forecaster to lag α of source variable i when predicting target variable j across the full forecasting horizon. This multiplicative construction combines an encoder-side source–lag structural prior with Jacobian-based local predictive sensitivity, so that high scores tend to arise when both terms are simultaneously large. Accordingly, Y^{causal} should be interpreted as a continuous lag-resolved Granger-style predictive dependence score, rather than a formal proof of structural causal identifiability.

For a compact pairwise edge score, we aggregate the lag profile by a normalized lag-group norm:

$$C_{j\leftarrow i} = s_{j,i}^{dec} \frac{1}{\sqrt{L}} \| Y^{\mathrm{causal}}_{j\leftarrow i,:} \|_2, \overline{C}_{j\leftarrow i} = \frac{C_{j\leftarrow i}}{\max\limits_{u\neq v} C_{u\leftarrow v} + \varepsilon}, \overline{C} \in \mathbb{R}^{M\times M}. \tag{15}$$

Here, $s_{j,i}^{\mathrm{dec}}$ is the decoder-induced variable-level readout preference defined in Section 3.2. Both Y^{causal} and $\overline{C}$ are continuous predictive-

dependence scores rather than binary graphs. This is intentional: the continuous form preserves graded evidence from encoder-side structural retention and Jacobian-based local sensitivity, and is directly used for ranking-based evaluation such as AUROC and AUPRC. The unnormalized statistic $C_{j\leftarrow i}$ is used for significance testing, whereas the normalized score $\overline{C}$ is used for visualization and cross-edge comparability.

When graph-structure recovery is required, we further convert $\overline{C}$ into a binary adjacency matrix through edge-wise significance filtering rather than a fixed raw-score threshold. Specifically, for each candidate edge $i\rightarrow j$, we generate B_{surr} source-wise surrogate samples by circularly shifting the source trajectory x_i with random offsets larger than the maximum lag L, while keeping the remaining variables unchanged. This procedure preserves the marginal distribution and temporal autocorrelation structure of the source variable to a large extent while destroying its temporal alignment with the target, thereby forming a permutation-based surrogate test for estimating the null distribution of edge scores (Ramdas et al., 2023). Let $\overline{C}_{j\leftarrow i}^{(b)}$ denote the corresponding surrogate edge score for the b-th null sample, and $\mathbf{1}(\cdot)$ denotes the indicator function, which returns 1 if the condition holds and 0 otherwise. We then compute the empirical one-sided p-value as:

$$p_{j\leftarrow i}=\frac{1+\sum_{b=1}^{B_{\text{surr}}}\mathbf{1}\left(\overline{C}_{j\leftarrow i}^{(b)}\geq\overline{C}_{j\leftarrow i}\right)}{B_{\text{surr}}+1} \tag{16}$$

Since all directed edges are tested simultaneously, we further apply the Benjamini–Hochberg false discovery rate correction (Benjamini and Hochberg, 1995) to obtain adjusted p-values $p_{j\leftarrow i}^{adj}$. The final recovered binary graph is defined as:

$$\mathrm{A}_{\mathrm{j\leftarrow i}}=1\left(\mathrm{p}_{\mathrm{j\leftarrow i}}^{\mathrm{adj}}<\mathrm{a}\right),\mathrm{A}\in\{0,1\}^{\mathrm{M}\times\mathrm{M}} \tag{17}$$

where α is the significance level and the concrete settings of the significance-filtering protocol, including α and B_{surr}, are given in Section 4.2.3. The recovered binary graph is used only for graph-recovery evaluation metrics such as SHD, Precision, Recall, and F1. Hence, the graph recovery step should be understood as a statistically calibrated evaluation-time post-processing procedure built on top of the continuous Granger-style predictive-dependence score.

This significance-filtering strategy improves the statistical reliability of edge selection and makes graph recovery more robust than direct thresholding on raw sensitivity scores. The recovered graph is therefore intended to provide Granger-style causal evidence for evaluation and interpretation, rather than as a formal claim of structural causality. For clarity, Fig. 3 consolidates the complete training-to-analysis procedure described above.

4. Experiments

In this section, we detail the experimental contents and results for our model applied to both time series forecasting and causal analysis tasks. In the time series forecasting experiments, we compare MKJR with other mainstream baseline models specifically designed for time series forecasting tasks. Likewise, in the causal analysis experiments, we evaluate our model against existing causal analysis models to demonstrate its effectiveness in causal discovery tasks.

In the two major experiments presented here, we introduce the data sets utilized, experimental environment configurations, settings for each experiment and the final experimental results and discussions. Additionally, after each comparative experiment, we present the results of our ablation studies to evaluate the contributions of each component of the model to specific tasks.

4.1. Time series forecasting experiments

This section evaluates the forecasting performance of MKJR under a unified multivariate long-term time series forecasting protocol. Beyond reporting predictive accuracy, we explicitly separate forecasting results, computational efficiency, and component-wise robustness analysis. This organization is necessary for the current version of MKJR because the final model integrates a KAN-based encoder-decoder, asymmetric local/global gating, a heterogeneous bidirectional dual-stream Mamba backbone, and a Jacobian-based causal regularizer. Therefore, the forecasting experiments are designed not only to verify whether the full model achieves lower prediction errors, but also to clarify which gains come from structural retention, gated preconditioning, cross-variable modeling, and gradient-level regularization.

4.1.1. Datasets

We evaluate MKJR on widely used multivariate long-term time series forecasting benchmarks covering power-system, traffic, and meteorological scenarios. These benchmarks were selected because they span different sampling granularities, variable dimensionalities, and degrees of periodicity, non-stationarity, and cross-variable dependence.

The ETT benchmark includes four subsets, ETTh1, ETTh2, ETTm1, and ETTm2, each containing 7 variables related to electricity transformer load and oil temperature. These datasets are standard benchmarks for evaluating periodic industrial dynamics under different sampling frequencies. The Electricity dataset records hourly electricity load series for 321 clients and is widely used to evaluate multivariate load forecasting under pronounced distribution shifts. The Traffic dataset contains 862 channels of hourly traffic measurements and therefore provides a challenging testbed for high-dimensional cross-variable dependency modeling. The Weather dataset contains 21 meteorological variables sampled every 10 min and is suitable for assessing forecasting under mixed periodic and trend-dominated temporal patterns.

For completeness, the basic statistics of the forecasting datasets are summarized in Table 1.

4.1.2. Experimental environment

The forecasting experiments were conducted under a unified Linux-based environment. Model training was performed on a server equipped with Intel(R) Xeon(R) Platinum 8352V CPUs and four NVIDIA RTX 4090 GPUs (24 GB each), while evaluation and inference latency measurement were conducted on an NVIDIA V100 GPU. The operating system was Ubuntu 22.04 LTS, and the CUDA environment was based on versions 11.8/12.4. The detailed hardware configuration is summarized in Table 2.

4.1.3. Baselines

We compare MKJR with ten representative forecasting baselines from different modeling families, including Transformer-based models, Mamba-based models, KAN-based models, CNN-based models, and linear models. Specifically, the compared methods include iTransformer (Li et al., 2023a), Crossformer (Zhang and Yan, 2023), TimeMachine (Ahamed and Cheng, 2024), S-Mamba (Wang et al., 2025), AdaKAN (Wu et al., 2025), TimeKAN (Huang et al., 2025), SCINet (Liu et al., 2022), TimesNet (Wu et al., 2022), DLinear (Zeng et al., 2023), and RLinear (Li

Table 1
The relevant statistical data of the selected datasets.

Dataset	Channels	Timesteps	Granularity
ETTh1, ETTh2	7	17,420	1 h
ETTm1, ETTm2	7	69,680	15 min
Electricity	321	26,304	1 h
Traffic	862	17,544	1 h
Weather	21	52,696	10 min

Table 2
The environment of experiments.

Component	Description
System	Ubuntu 22.04 LTS Linux
CPU	Intel(R) Xeon(R) Platinum 8352V CPU
Memory Size	120 GB
Training GPU	NVIDIA RTX 4090(24 GB) * 4
Evaluation GPU	NVIDIA V100 GPU
CUDA Version	11.8/12.4

et al., 2023b).

These baselines were selected for three reasons. First, they cover the dominant architectural paradigms currently used in long-term forecasting, including attention-based sequence modeling, selective state-space modeling, interpretable spline-based approximation, convolutional temporal modeling, and strong linear baselines. Second, they include both highly expressive models and lightweight competitive models, which is important for judging whether MKJR improves accuracy through meaningful modeling gains rather than through excessive model capacity alone. Third, several of these methods have demonstrated strong performance on the same benchmarks adopted in this study, making them suitable reference points for a fair comparison.

4.1.4. Experimental details

For all forecasting models on all datasets, we use a unified look-back window length $L = 96$ and prediction horizons $T \in \{96, 192, 336, 720\}$. For patch-based models, we follow the standard setting with patch length $P = \frac{L}{4}$ and stride $S = \frac{P}{2}$, while preserving the original patch granularity of methods whose official implementations are sensitive to patch design. The hidden dimension is selected from $\{32, 64, 128, 256, 512\}$ according to validation performance. The current implementation of MKJR is trained for at most 100 epochs with early stopping of patience 5, using the Adam optimizer. The batch size is selected from $\{16, 32, 64, 128, 256, 512\}$ according to validation stability and GPU memory usage. We perform learning-rate search over $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$.

For MKJR, the KAN hyperparameters follow the KAN4TSF setting for fairness. The grid interval number is set to 5 and the B-spline order is set to 3. Unless otherwise stated, the encoder-decoder widths are set to $(N_1, N_2) = (256, 128)$, and the decoder mirrors the encoder structure. To regularize the KAN encoder-decoder, we apply spline dropout with rate 0.05 inside each KAN layer and standard dropout with rate 0.10 between successive KAN stages.

MKJR additionally contains asymmetric local/global gating before the dual-stream Mamba backbone. In the default forecasting configuration, the local descriptor extractor $\Phi_{\text{spec}}(\cdot)$ is instantiated as a lightweight variable-wise temporal descriptor module with kernel size 3, while the global fusion mapping $\Gamma(\cdot)$ is implemented as a single linear projection over the concatenated compressed token and pooled context. Each stream contains one forward and one reversed Mamba block. The detail stream operates on the N_1-dimensional tokens without an identity residual shortcut, whereas the global stream operates on the N_2-dimensional compressed tokens with an additional residual path. The decoder-side variable-level readout preference used later in causal scoring is induced post hoc from the trained KAN decoder rather than from an additional standalone routing matrix.

For training-time Jacobian regularization, we use the matrix-free random-projection objective described in Section 3.4. Unless otherwise stated, the probe budget is fixed to $n_{\text{proj}} = 4$, and the outer coefficients are set to $\lambda_{\text{sens}} = 0.10$ and $\lambda_{\text{str}} = 0.10$. These values are treated as default settings rather than final optimal choices, and a dedicated sensitivity study is reported later. Following common practice in multivariate forecasting under temporal distribution shift, we adopt RevIN as the default preprocessing strategy and examine its necessity separately in the ablation study.

We use MSE and MAE as the primary forecasting metrics. To improve reliability, all forecasting results reported in the following subsections are presented as mean $\pm$ standard deviation over five runs with different random seeds $\{2026, 2027, 2028, 2029, 2030\}$. Besides predictive accuracy, we also record the number of parameters, FLOPs, training time per epoch and inference latency under a unified measurement protocol.

4.1.5. Results

The forecasting results are summarized in Tables 3 and 4. Overall, MKJR achieves consistently competitive performance across diverse prediction horizons and dataset characteristics, indicating that the current architecture generalizes well across periodic, high-dimensional, and non-stationary multivariate series.

The gains of MKJR should not be attributed to a single architectural trick. Instead, they come from the joint effect of four design choices. First, the KAN encoder-decoder provides a structured nonlinear mapping that preserves variable-wise temporal signatures before cross-variable dependency modeling. Second, the asymmetric local/global gating mechanism improves the conditioning of the two streams by emphasizing fine-grained temporal signatures in the detail stream and coarse cross-variable context in the global stream. Third, the bidirectional dual-stream Mamba backbone models predictive interactions across variable tokens from complementary directions. Fourth, Jacobian-based regularization improves training stability and reduces unstable local sensitivity during optimization.

Dataset-wise, the improvements are expected to be heterogeneous. On ETT-type datasets with relatively clearer periodicity, the structural retention effect of the KAN encoder and local gating is more prominent. On Electricity, which exhibits stronger volatility and distribution shift, RevIN and Jacobian-based regularization contribute more to stabilizing optimization and improving generalization. On Traffic, which contains 862 variables and stronger cross-channel interactions, the benefits of cross-variable modeling and global gating are expected to be more visible. This tendency is also consistent with recent Mamba-based forecasting studies showing that variable-mixing improvements become more noticeable on high-dimensional datasets such as Traffic.

To ensure a fair main comparison, all methods are evaluated under the same forecasting metrics and data protocol. For baseline methods with publicly available and reproducible implementations, we rerun the experiments under a unified environment and report the mean and standard deviation over multiple runs. Results without standard deviation are reported from the original papers.

Although AdaKAN, S-Mamba, and TimeMachine are competitive on a few datasets, MKJR achieves better overall and more stable performance. This indicates that the integration of KAN and Mamba is beneficial, as it combines strong nonlinear representation with effective long-range dependency modeling. Transformer-based models are competitive for short-term forecasting, but their performance drops at longer horizons. CNN-based and linear models perform well in specific settings, yet they are less robust across different temporal patterns and granularities. Overall, the comparative experimental results in different scenarios and datasets show that MKJR outperforms other time series forecasting models, especially for long-term forecasting tasks.

To further assess whether the observed improvements are robust across random seeds, we additionally compare MKJR with TimeMachine using paired t-tests on the five run-wise MSE values for each dataset-horizon setting. The detailed results are reported in Table 5.

As shown in Table 5, MKJR consistently achieves lower mean MSE than TimeMachine across all reported settings. Moreover, all paired t-test p-values are below 0.05, suggesting that the observed improvements are unlikely to be caused solely by random seed variation.

4.1.6. Ablation study

We conduct an ablation study to evaluate the contribution of the major components of MKJR to forecasting performance. Specifically, we separately remove the KAN layers, the bidirectional dual-stream

Table 3
MSE results for long-term prediction tasks.

Dataset		Our Model	KAN-based Model		Transformer-based Model		Mamba-based Model		CNN-based Model		Linear-based Model	
	T	MKJR	AdaKAN (2025)	TimeKAN (2025)	iTrans former (2024)	Crossformer (2023)	TimeMachine (2024)	S-Mamba (2025)	SCINet (2022)	TimesNet (2022)	DLinear (2023)	RLinear (2023)
		MSE	MSE	MSE	MSE	MSE	MSE	MSE	MSE	MSE	MSE	MSE
ETTm1	96	**0.309** ± 0.009	0.312	0.322 ± 0.012	0.334 ± 0.014	0.404 ± 0.015	0.317 ± 0.012	0.333 ± 0.014	0.418 ± 0.016	0.338 ± 0.014	0.353 ± 0.014	0.355
	192	**0.331** ± 0.012	0.340	0.357 ± 0.016	0.377 ± 0.018	0.450 ± 0.020	0.357 ± 0.016	0.375 ± 0.018	0.439 ± 0.019	0.374 ± 0.017	0.389 ± 0.018	0.391
	336	**0.365** ± 0.011	0.371	0.382 ± 0.018	0.426 ± 0.017	0.532 ± 0.025	0.379 ± 0.018	0.408 ± 0.017	0.490 ± 0.018	0.410 ± 0.016	0.421 ± 0.015	0.424
	720	**0.423** ± 0.013	0.431	0.445 ± 0.019	0.491 ± 0.019	0.666 ± 0.027	0.445 ± 0.019	0.475 ± 0.015	0.595 ± 0.022	0.478 ± 0.014	0.484 ± 0.015	0.487
	avg	**0.357**	0.364	0.376	0.407	0.513	0.375	0.398	0.485	0.400	0.412	0.414
ETTm2	96	**0.165** ± 0.006	0.169	0.174 ± 0.008	0.180 ± 0.009	0.287 ± 0.012	0.175 ± 0.008	0.179 ± 0.008	0.286 ± 0.011	0.187 ± 0.008	0.182 ± 0.008	0.182
	192	**0.231** ± 0.009	0.232	0.239 ± 0.012	0.250 ± 0.011	0.414 ± 0.016	0.239 ± 0.010	0.250 ± 0.011	0.399 ± 0.019	0.249 ± 0.012	0.257 ± 0.013	0.246
	336	**0.269** ± 0.010	0.277	0.301 ± 0.015	0.311 ± 0.012	0.597 ± 0.021	0.287 ± 0.011	0.312 ± 0.014	0.637 ± 0.026	0.321 ± 0.012	0.318 ± 0.012	0.307
	720	**0.366** ± 0.014	0.372	0.395 ± 0.019	0.412 ± 0.016	1.730 ± 0.039	0.371 ± 0.017	0.411 ± 0.016	0.960 ± 0.032	0.408 ± 0.022	0.426 ± 0.018	0.407
	avg	**0.258**	0.262	0.277	0.288	0.757	0.268	0.288	0.571	0.291	0.296	0.286
ETTh1	96	**0.352** ± 0.008	0.374	0.367 ± 0.012	0.386 ± 0.018	0.423 ± 0.012	0.364 ± 0.009	0.386 ± 0.019	0.654 ± 0.025	0.384 ± 0.018	0.387 ± 0.019	0.386
	192	**0.407** ± 0.013	0.414	0.414 ± 0.016	0.441 ± 0.019	0.471 ± 0.017	0.415 ± 0.016	0.443 ± 0.019	0.719 ± 0.029	0.436 ± 0.019	0.439 ± 0.019	0.437
	336	**0.414** ± 0.016	0.443	0.445 ± 0.019	0.487 ± 0.017	0.570 ± 0.020	0.429 ± 0.018	0.489 ± 0.021	0.778 ± 0.030	0.491 ± 0.020	0.493 ± 0.025	0.479
	720	**0.432** ± 0.016	0.452	0.444 ± 0.019	0.503 ± 0.021	0.653 ± 0.021	0.458 ± 0.021	0.502 ± 0.022	0.836 ± 0.033	0.521 ± 0.025	0.490 ± 0.017	0.481
	avg	**0.401**	0.421	0.417	0.454	0.529	0.417	0.455	0.747	0.458	0.452	0.446
ETTh2	96	**0.227** ± 0.006	0.313	0.290 ± 0.011	0.297 ± 0.008	0.745 ± 0.036	0.275 ± 0.011	0.296 ± 0.009	0.707 ± 0.035	0.340 ± 0.016	0.305 ± 0.012	0.288
	192	**0.285** ± 0.013	0.397	0.375 ± 0.018	0.380 ± 0.018	0.877 ± 0.034	0.349 ± 0.013	0.376 ± 0.018	0.860 ± 0.037	0.402 ± 0.018	0.424 ± 0.025	0.374
	336	**0.338** ± 0.012	0.441	0.423 ± 0.019	0.428 ± 0.017	1.043 ± 0.040	0.340 ± 0.015	0.424 ± 0.016	1.000 ± 0.043	0.452 ± 0.021	0.456 ± 0.021	0.415
	720	**0.405** ± 0.015	0.453	0.443 ± 0.019	0.427 ± 0.020	1.104 ± 0.036	0.411 ± 0.016	0.426 ± 0.025	1.249 ± 0.049	0.462 ± 0.025	0.476 ± 0.023	0.420
	avg	**0.314**	0.401	0.383	0.383	0.942	0.344	0.381	0.954	0.414	0.415	0.374
ECL	96	**0.139** ± 0.005	0.164	0.174 ± 0.007	0.148 ± 0.016	0.219 ± 0.014	0.142 ± 0.011	**0.139** ± 0.007	0.247 ± 0.013	0.168 ± 0.011	0.206 ± 0.011	0.201
	192	**0.152** ± 0.006	0.178	0.182 ± 0.0010	0.162 ± 0.017	0.231 ± 0.016	0.158 ± 0.009	0.156 ± 0.009	0.257 ± 0.018	0.184 ± 0.013	0.206 ± 0.007	0.201
	336	**0.163** ± 0.005	0.192	0.197 ± 0.006	0.178 ± 0.011	0.246 ± 0.011	0.172 ± 0.015	0.172 ± 0.010	0.269 ± 0.016	0.198 ± 0.010	0.220 ± 0.010	0.215
	720	**0.193** ± 0.008	0.277	0.236 ± 0.011	0.225 ± 0.012	0.280 ± 0.013	0.207 ± 0.014	0.204 ± 0.013	0.299 ± 0.011	0.220 ± 0.013	0.252 ± 0.009	0.257
	avg	**0.162**	0.190	0.197	0.178	0.244	0.170	0.170 ± 0.010	0.268	0.192	0.221	0.219
Traffic	96	**0.371** ± 0.013	0.527	0.575 ± 0.020	0.395 ± 0.019	0.522 ± 0.027	0.397 ± 0.019	0.382 ± 0.018	0.788 ± 0.034	0.593 ± 0.020	0.647 ± 0.027	0.649
	192	**0.387** ± 0.015	0.535	0.550 ± 0.023	0.417 ± 0.016	0.530 ± 0.024	0.417 ± 0.016	0.396 ± 0.019	0.789 ± 0.033	0.617 ± 0.028	0.596 ± 0.024	0.601
	336	**0.401** ± 0.016	0.541	0.576 ± 0.028	0.433 ± 0.019	0.558 ± 0.020	0.433 ± 0.019	0.417 ± 0.017	0.797 ± 0.032	0.629 ± 0.025	0.601 ± 0.029	0.609
	720	0.426 ± 0.019	0.560	0.602 ± 0.026	0.467 ± 0.021	0.589 ± 0.025	0.467 ± 0.022	**0.406** ± 0.020	0.841 ± 0.039	0.640 ± 0.027	0.642 ± 0.029	0.647
	avg	**0.396**	0.540	0.576	0.428	0.550	0.429	0.414	0.804	0.620	0.622	0.627
Weather	96	**0.148** ± 0.004	0.150	0.162 ± 0.005	0.174 ± 0.009	0.158 ± 0.006	0.164 ± 0.012	0.165 ± 0.008	0.221 ± 0.017	0.172 ± 0.009	0.196 ± 0.007	0.192
	192	**0.192** ± 0.006	**0.192**	0.207 ± 0.007	0.221 ± 0.012	0.206 ± 0.011	0.211 ± 0.013	0.214 ± 0.014	0.261 ± 0.013	0.219 ± 0.010	0.241 ± 0.014	0.240
	336	**0.237** ± 0.009	0.242	0.263 ± 0.010	0.278 ± 0.013	0.272 ± 0.013	0.256 ± 0.013	0.274 ± 0.011	0.309 ± 0.015	0.280 ± 0.014	0.292 ± 0.012	0.292
	720	**0.312** ± 0.011	**0.312**	0.338 ± 0.014	0.358 ± 0.016	0.398 ± 0.019	0.342 ± 0.017	0.350 ± 0.014	0.377 ± 0.018	0.365± 0.019	0.363 ± 0.018	0.364
	avg	**0.222**	0.224	0.242	0.258	0.259	0.244	0.251	0.292	0.259	0.273	0.272

Table 4
MAE results for long-term prediction tasks.

Dataset		Our Model	KAN-based Model		Transformer-based Model		Mamba-based Model		CNN-based Model		Linear-based Model	
	T	MKJR	AdaKAN (2025)	TimeKAN (2025)	iTrans former (2024)	Crossformer (2023)	TimeMachine (2024)	S-Mamba (2025)	SCINet (2022)	TimesNet (2022)	DLinear (2023)	RLinear (2023)
		MAE	MAE	MAE	MAE	MAE	MAE	MAE	MAE	MAE	MAE	MAE
ETTm1	96	**0.348** ± 0.011	0.355	0.361 ± 0.013	0.368 ± 0.015	0.426 ± 0.017	0.355 ± 0.014	0.368 ± 0.015	0.418 ± 0.016	0.375 ± 0.018	0.374 ± 0.017	0.376
	192	**0.357** ± 0.014	0.361	0.383 ± 0.018	0.391 ± 0.019	0.451 ± 0.021	0.378 ± 0.018	0.390 ± 0.019	0.439 ± 0.019	0.387 ± 0.018	0.391 ± 0.016	0.392
	336	**0.379** ± 0.013	0.386	0.401 ± 0.015	0.420 ± 0.019	0.515 ± 0.023	0.399 ± 0.017	0.413 ± 0.016	0.490 ± 0.021	0.411 ± 0.016	0.413 ± 0.017	0.415
	720	**0.417** ± 0.016	0.425	0.435 ± 0.019	0.459 ± 0.021	0.589 ± 0.026	0.436 ± 0.019	0.448 ± 0.019	0.595 ± 0.024	0.450 ± 0.021	0.448 ± 0.019	0.450
	avg	**0.375**	0.382	0.395	0.410	0.495	0.392	0.405	0.485	0.406	0.407	0.408
ETTm2	96	**0.246** ± 0.009	0.253	0.255 ± 0.011	0.264 ± 0.012	0.366 ± 0.015	0.256 ± 0.013	0.263 ± 0.010	0.286 ± 0.011	0.267 ± 0.012	0.264 ± 0.011	0.265
	192	**0.271** ± 0.010	0.298	0.299 ± 0.013	0.309 ± 0.016	0.492 ± 0.023	0.299 ± 0.012	0.309 ± 0.018	0.399 ± 0.019	0.309 ± 0.017	0.315 ± 0.016	0.304
	336	**0.319** ± 0.012	0.335	0.340 ± 0.015	0.348 ± 0.012	0.542 ± 0.022	0.332 ± 0.014	0.349 ± 0.016	0.637 ± 0.018	0.351 ± 0.016	0.353 ± 0.014	0.342
	720	**0.353** ± 0.018	0.383	0.396 ± 0.019	0.407 ± 0.015	1.042 ± 0.019	0.385 ± 0.018	0.406 ± 0.009	0.960 ± 0.034	0.403 ± 0.016	0.419 ± 0.016	0.398
	avg	**0.297**	0.317	0.322	0.332	0.611	0.318	0.332	0.571	0.333	0.338	0.327
ETTh1	96	**0.364** ± 0.011	0.393	0.395 ± 0.019	0.405 ± 0.020	0.448 ± 0.019	0.387 ± 0.019	0.405 ± 0.019	0.654 ± 0.026	0.402 ± 0.019	0.406 ± 0.018	0.395
	192	**0.374** ± 0.013	0.424	0.420 ± 0.015	0.436 ± 0.019	0.474 ± 0.014	0.416 ± 0.016	0.437 ± 0.015	0.719 ± 0.018	0.429 ± 0.021	0.435 ± 0.019	0.424
	336	**0.401** ± 0.015	0.432	0.435 ± 0.019	0.458 ± 0.021	0.546 ± 0.018	0.421 ± 0.017	0.468 ± 0.019	0.778 ± 0.027	0.469 ± 0.018	0.457 ± 0.021	0.446
	720	**0.433** ± 0.014	0.455	0.459 ± 0.021	0.491 ± 0.015	0.621 ± 0.021	0.453 ± 0.021	0.489 ± 0.017	0.836 ± 0.032	0.500 ± 0.020	0.478 ± 0.022	0.470
	avg	**0.393**	0.426	0.427	0.448	0.522	0.419	0.450	0.747	0.450	0.444	0.434
ETTh2	96	**0.299** ± 0.009	0.334	0.340 ± 0.015	0.349 ± 0.015	0.584 ± 0.019	0.334 ± 0.014	0.348 ± 0.015	0.707 ± 0.025	0.374 ± 0.018	0.352 ± 0.016	0.338
	192	**0.329** ± 0.012	0.391	0.392 ± 0.016	0.400 ± 0.019	0.656 ± 0.024	0.381 ± 0.018	0.396 ± 0.016	0.860 ± 0.033	0.414 ± 0.016	0.439 ± 0.019	0.390
	336	**0.374** ± 0.009	0.395	0.435 ± 0.019	0.432 ± 0.018	0.731 ± 0.021	0.381 ± 0.019	0.431 ± 0.018	1.000 ± 0.028	0.452 ± 0.021	0.473 ± 0.017	0.426
	720	**0.421** ± 0.015	0.433	0.449 ± 0.017	0.445 ± 0.019	0.763 ± 0.026	0.433 ± 0.018	0.444 ± 0.019	1.249 ± 0.036	0.468 ± 0.019	0.493 ± 0.020	0.440
	avg	**0.356**	0.388	0.404	0.407	0.684	0.382	0.405	0.954	0.427	0.439	0.399
ECL	96	**0.231** ± 0.007	0.261	0.266 ± 0.012	0.240 ± 0.015	0.314 ± 0.021	0.236 ± 0.013	0.235 ± 0.018	0.247 ± 0.015	0.272 ± 0.016	0.288 ± 0.015	0.281
	192	**0.247** ± 0.011	0.273	0.273 ± 0.015	0.253 ± 0.019	0.322 ± 0.018	0.250 ± 0.018	0.255 ± 0.017	0.257 ± 0.019	0.289 ± 0.013	0.290 ± 0.015	0.283
	336	**0.262** ± 0.009	0.282	0.286 ± 0.016	0.269 ± 0.017	0.337 ± 0.017	0.268 ± 0.016	0.272 ± 0.019	0.269 ± 0.021	0.300 ± 0.015	0.305 ± 0.012	0.298
	720	**0.289** ± 0.015	0.316	0.320 ± 0.012	0.317 ± 0.020	0.363 ± 0.022	0.298 ± 0.011	0.298 ± 0.014	0.299 ± 0.020	0.320 ± 0.012	0.337 ± 0.014	0.331
	avg	**0.257** ± 0.011	0.283	0.286 ± 0.014	0.270 ± 0.018	0.334 ± 0.020	0.263 ± 0.018	0.265 ± 0.017	0.268	0.295 ± 0.014	0.305 ± 0.014	0.298
Traffic	96	**0.259** ± 0.013	0.338	0.378 ± 0.018	0.268 ± 0.015	0.290 ± 0.017	0.268 ± 0.015	0.261 ± 0.011	0.788 ± 0.026	0.321 ± 0.012	0.384 ± 0.018	0.389
	192	**0.265** ± 0.018	0.344	0.362 ± 0.015	0.276 ± 0.013	0.293 ± 0.015	0.274 ± 0.019	0.267 ± 0.015	0.789± 0.024	0.336 ± 0.014	0.359 ± 0.016	0.366
	336	**0.271** ± 0.016	0.349	0.378 ± 0.013	0.283 ± 0.016	0.305 ± 0.018	0.281 ± 0.012	0.276 ± 0.016	0.797 ± 0.028	0.336 ± 0.013	0.361 ± 0.015	0.369
	720	**0.282** ± 0.011	0.357	0.380 ± 0.018	0.302 ± 0.019	0.328 ± 0.012	0.300 ± 0.013	0.300 ± 0.018	0.841 ± 0.019	0.350 ± 0.015	0.381 ± 0.018	0.387
	avg	**0.269**	0.347	0.375	0.282	0.304	0.281	0.276	0.804	0.336	0.371	0.378
Weather	96	0.203 ± 0.013	**0.201**	0.208 ± 0.014	0.214 ± 0.015	0.230 ± 0.012	0.208 ± 0.010	0.210 ± 0.012	0.221 ± 0.017	0.220 ± 0.011	0.235 ± 0.019	0.232
	192	**0.241** ± 0.010	**0.241**	0.249 ± 0.016	0.254 ± 0.017	0.277 ± 0.017	0.250 ± 0.015	0.252 ± 0.019	0.261 ± 0.018	0.261 ± 0.014	0.271 ± 0.017	0.271
	336	**0.284** ± 0.012	0.286	0.290 ± 0.015	0.296 ± 0.012	0.335 ± 0.014	0.290 ± 0.014	0.297 ± 0.014	0.309 ± 0.021	0.306 ± 0.015	0.306 ± 0.018	0.307
	720	**0.334** ± 0.014	0.338	0.340 ± 0.012	0.349 ± 0.018	0.418 ± 0.016	0.343 ± 0.017	0.345 ± 0.020	0.377 ± 0.018	0.359 ± 0.016	0.353 ± 0.015	0.353
	avg	**0.266**	**0.266**	0.272	0.278	0.315	0.272	0.276	0.292	0.287	0.291	0.291

Table 5
Statistical comparison between MKJR and TimeMachine across forecasting settings (MSE).

Dataset	Horizon	MKJR (mean ± std)	TimeMachine (mean ± std)	p-value (paired *t*-test, two-sided)
ETTm1	96	0.309 ± 0.009	0.317 ± 0.012	0.0040
	192	0.331 ± 0.012	0.357 ± 0.016	0.0001
	336	0.365 ± 0.011	0.379 ± 0.018	0.0112
	720	0.423 ± 0.013	0.445 ± 0.019	0.0012
Traffic	96	0.371 ± 0.013	0.397 ± 0.019	0.0006
	192	0.387 ± 0.015	0.417 ± 0.016	2.40e-07
	336	0.401 ± 0.016	0.433 ± 0.019	1.84e-05
	720	0.426 ± 0.019	0.467 ± 0.022	6.87e-06
Weather	96	0.148 ± 0.004	0.164 ± 0.012	0.0111
	192	0.192 ± 0.006	0.211 ± 0.013	0.0036
	336	0.237 ± 0.009	0.256 ± 0.013	0.0004
	720	0.312 ± 0.011	0.342 ± 0.017	0.0004

architecture, and the Jacobian-based regularizer, and examine their effects on ETTm1, Electricity, and Traffic. These experiments help identify the key components of the framework and clarify their roles under different data characteristics.

The results reveal dataset-dependent contributions of the three components. On ETTm1, which exhibits clear periodicity and relatively low volatility, removing the KAN embedding and mapping layers causes a substantial performance drop. This suggests that KAN is effective in capturing both the intrinsic nonlinear structure and the stable periodic dynamics of the data. As further illustrated in Fig. 4, replacing the original MLP-based embedding and mapping with KAN layers enables MKJR to better exploit periodic patterns and long-term trends.

On Electricity, which is more volatile and non-stationary, removing the Jacobian-based regularizer leads to a clear degradation in forecasting performance. As shown in Fig. 5, this indicates that the regularizer plays an important role in stabilizing prediction and improving generalization under highly variable conditions.

On Traffic, where complex interactions exist across multiple time-series channels, removing the bidirectional dual-stream architecture significantly reduces prediction accuracy. Fig. 6 shows that this module is important for capturing bidirectional temporal interactions and broader contextual dependencies, which are crucial for modeling complex nonlinear relationships in real-world traffic data.

To avoid fragmented reporting while still isolating the contribution of each major component, we perform a unified one-factor-at-a-time ablation study for forecasting, as summarized in Table 6. Specifically, we remove the KAN module, the Mamba backbone, the local gate G_{loc}, the global gate G_{glo}, the sensitivity regularizer R_{sens}, the structured sparsity term R_{str}, and the whole Jacobian-based objective, respectively. Here, w/o Jacobian means removing both R_{sens} and R_{str} simultaneously.

Several observations can be made. First, removing KAN or Mamba causes the most notable degradation in forecasting performance, confirming that nonlinear variable-wise temporal representation and long-range dependency modeling remain the two main drivers of predictive accuracy. Second, removing the asymmetric gating modules also leads to clear performance drops. In particular, removing G_{loc} hurts ETTm1 more, which is consistent with the role of local gating in preserving fine-grained temporal signatures, while removing G_{glo} affects Traffic more, indicating the importance of coarse global context in high-dimensional cross-variable forecasting. Third, removing R_{sens} or R_{str} still consistently degrades performance, but the magnitude is smaller than that of removing KAN, Mamba, or the major gating modules. This suggests that the Jacobian-related regularizers mainly play a supportive yet non-negligible role in forecasting by improving optimization stability and structured selectivity. Finally, removing the whole Jacobian objective causes a larger drop than removing either R_{sens} or R_{str} alone, showing that the two regularizers work complementarily during training.

We additionally compare several recent normalization strategies for non-stationary forecasting, including RevIN, Dish-TS (Fan et al., 2023), SAN (Liu et al., 2023b), SIN (Han et al., 2024a), and DDN (Dai et al., 2024). The results are summarized in Table 7. Although some dynamic normalization methods yield slightly better results on specific datasets, the performance differences remain relatively small. To ensure fair comparison with mainstream recent LTSF studies and to avoid over-attributing the gain to a stronger normalization module alone, we adopt RevIN as the default setting in the main experiments.

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4.1.7. Hyperparameter sensitivity analysis

We further examine the sensitivity of MKJR to several key hyperparameters, including the expert kernel size Φ_{spec}, the hidden-width configuration (N_1, N_2), the spline grid interval number, and the B-spline order. The results on ETTh1 and Weather are summarized in Table 8.

The results indicate that MKJR is relatively stable within a moderate hyperparameter range. For the expert kernel size, $\Phi_{\text{spec}} = 3$ yields the best overall performance on both ETTh1 and Weather, suggesting that a moderate receptive range is sufficient to balance local detail modeling and robustness. For the hidden-width setting, $(N_1, N_2) = (256, 128)$ achieves the best results, while both smaller and larger configurations lead to performance degradation, indicating that excessively weak or overly heavy hidden representations are both suboptimal. For the KAN spline settings, a grid interval number of 5 and a B-spline order of 3 provide the best performance, whereas either overly coarse or overly flexible spline parameterizations tend to degrade forecasting accuracy.

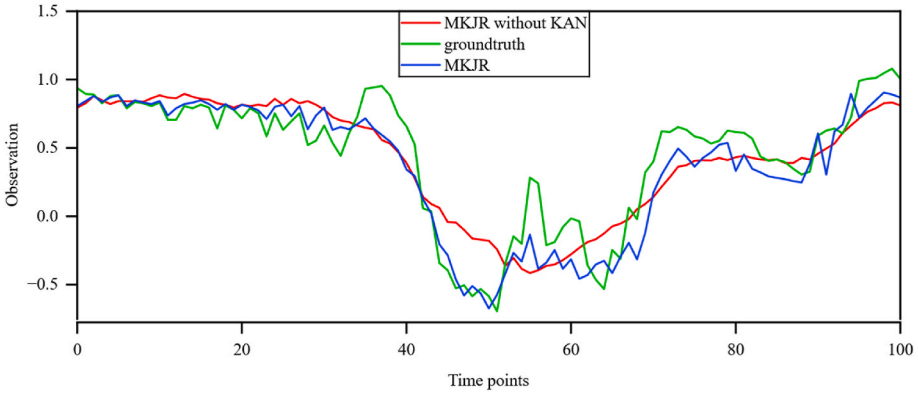


Fig. 4. KAN ablation on ETTm1 dataset.

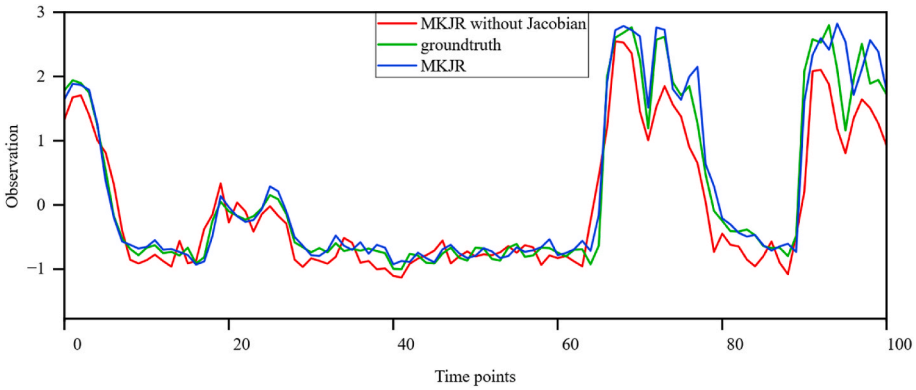


Fig. 5. Jacobian ablation on Electricity dataset.

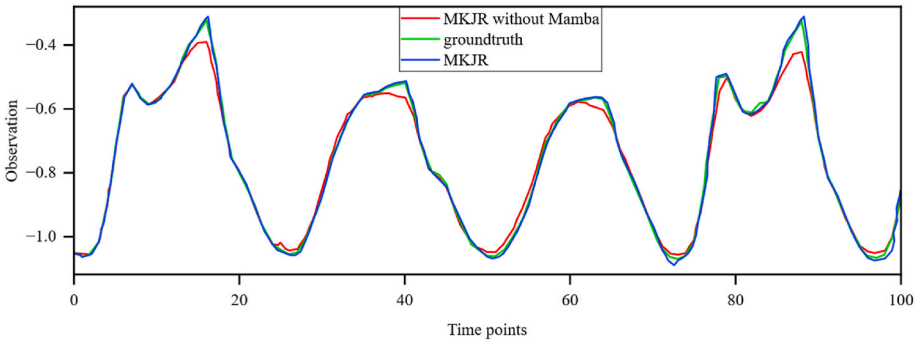


Fig. 6. Mamba ablation on Traffic dataset.

Table 6
Unified ablation study of MKJR for multivariate long-term forecasting (average over four horizons).

Variants	ETTm1 (Avg.)		Electricity (Avg.)		Traffic (Avg.)	
	MSE	MAE	MSE	MAE	MSE	MAE
w/o KAN	0.394	0.404	0.316	0.305	0.426	0.335
w/o Mamba	0.432	0.429	0.379	0.398	0.491	0.418
w/o G^{loc}	0.389	0.397	0.269	0.288	0.419	0.291
w/o G^{glo}	0.386	0.394	0.270	0.289	0.421	0.316
w/o R_{sens}	0.383	0.392	0.268	0.287	0.399	0.273
w/o R_{str}	0.384	0.393	0.267	0.286	0.401	0.270
w/o Jacobian	0.392	0.401	0.311	0.309	0.441	0.357
MKJR	**0.357**	**0.362**	**0.162**	**0.257**	**0.396**	**0.269**

Table 7
Comparison of normalization strategies for MKJR on non-stationary long-term forecasting benchmarks.

Normalization	RevIN	Dish-TS	SAN	SIN	DDN
	MSE	MSE	MSE	MSE	MSE
ETTh1	0.401	0.403	**0.395**	0.397	0.408
Weather	0.222	0.221	**0.215**	0.216	0.225

Based on these observations, we use $\Phi_{\text{spec}} = 3$, $(N_1, N_2) = (256, 128)$, a grid interval number of 5, and a B-spline order of 3 in the main experiments.

4.1.8. Complexity and efficiency analysis

In addition to forecasting accuracy, we analyze the computational characteristics of MKJR from the perspectives of model size, training efficiency and inference latency. This distinction is particularly important for MKJR because its Jacobian-based component affects optimization and interpretability more strongly than standard forward inference.

Table 8
Hyperparameter sensitivity analysis of MKJR on ETTh1 and Weather.

Φ_{spec} kernel size	1	2	3	4	5
	MSE	MSE	MSE	MSE	MSE
ETTh1	0.444	0.426	**0.401**	0.483	0.518
Weather	0.267	0.249	**0.222**	0.269	0.312
(N_1, N_2)	(128,64)	(256,64)	(256,128)	(512,128)	(512,256)
	MSE	MSE	MSE	MSE	MSE
ETTh1	0.567	0.526	**0.401**	0.419	0.463
Weather	0.352	0.283	**0.222**	0.269	0.294
Grid interval number	3	5	7	9	11
	MSE	MSE	MSE	MSE	MSE
ETTh1	0.462	**0.401**	0.429	0.459	0.564
Weather	0.353	**0.222**	0.237	0.307	0.390
B-spline order	2	3	4	5	6
	MSE	MSE	MSE	MSE	MSE
ETTh1	0.531	**0.401**	0.452	0.441	0.587
Weather	0.382	**0.222**	0.262	0.236	1.329

Fig. 7 summarizes the efficiency-accuracy trade-off of representative forecasting models on the Electricity dataset under the unified protocol described in Section 4.1.3, where the forecasting horizon is fixed to $T = 720$ and the input length is fixed to $L = 96$.

As shown in Fig. 7, MKJR achieves a favorable efficiency-performance trade-off among recent forecasting models. Although it is not the lightest architecture in terms of parameters or FLOPs, it remains clearly more compact than heavier baselines such as Crossformer and TimesNet, while achieving lower MAE. In terms of training and inference time, MKJR introduces a moderate practical overhead due to the combination of KAN representation learning and the shared dual-stream Mamba backbone, but it does not fall into the heaviest regime. Overall,

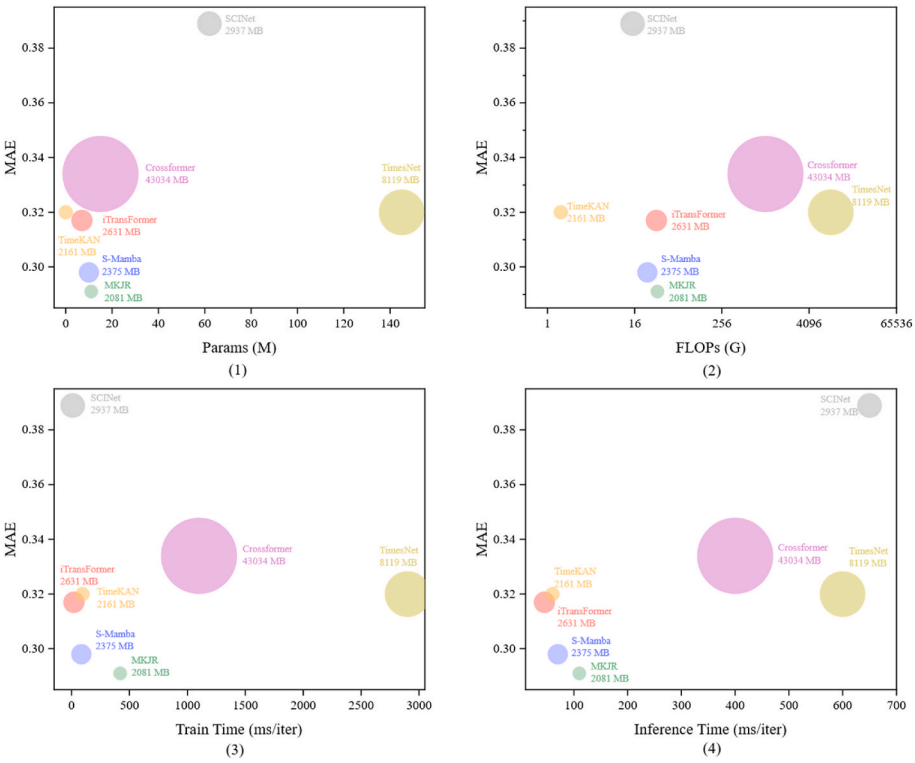


Fig. 7. Efficiency-accuracy trade-off of recent forecasting models on the Electricity dataset.

these results indicate that the Mamba + KAN design improves predictive performance without making the model excessively expensive in practice.

4.2. Causal analysis experiments

4.2.1. Datasets

In the causal analysis experiments (including ablation studies), we selected a total of 8 datasets for causal analysis and inference, covering not only standard synthetic and benchmark scenarios but also real industrial process datasets for practical causal validation. These datasets include.

(1) VAR Dataset: Simulated data with multivariate interactions generated using the Vector Autoregressive (VAR) model.
(2) Lorenz-96: A widely used dataset for studying climate dynamics and complex systems. It is a nonlinear dynamic system described by a set of Ordinary Differential Equations (ODEs).
(3) fMRI: Functional Magnetic Resonance Imaging (fMRI) data, an important source for studying brain functions and neural activity. It indirectly reflects neural activity by measuring blood oxygen level changes in different regions of the brain.
(4) DREAM-3: A benchmark dataset for causal inference, designed based on a challenge focused on gene regulatory networks. It describes the regulatory relationships between genes, typically in sparse directed graphs.
(5) CausalTime Dataset: A comprehensive dataset generated by (Cheng et al., 2023) for time series causal inference research. It includes sub-datasets such as AQI (Air Quality Index), Traffic and Medical information.
(6) Tennessee-Eastman (TE): A representative industrial process dataset widely used in process monitoring, fault diagnosis, and causal discovery research (McAvoy and Ye, 1994). It describes a complex chemical process with strong interactions among process variables and serves as an important benchmark for validating causal structure recovery in industrial systems.
(7) Ultra-processed Food (UF): A real-world industrial process dataset collected from an ultra-processed food production scenario (Menegozzo et al., 2020). Compared with simulated benchmarks, this dataset better reflects the complexity, noise, and structural heterogeneity of real industrial systems, and is therefore suitable for evaluating the practical effectiveness of causal analysis methods in industrial environments.
(8) Debutanizer (DE): A real industrial process dataset associated with a debutanizer column, which has been widely used in industrial process modeling and process analysis research (Fortuna et al., 2005). Owing to its realistic variable coupling and process dependencies, it provides an additional benchmark for validating causal analysis performance in practical industrial applications.

The environment configuration for the causal analysis experiment is the same as that for the time series forecasting experiments, as described in Table 2.

4.2.2. Baselines

We selected 9 representative causal as benchmark models, including TCDF (Nauta et al., 2019), eSRU (Khanna and Tan, 2019), cMLP, cLSTM (Tank et al., 2021), Rhino (Gong et al., 2022), JGC (Suryadiet al.2023), LOCAL (Cheng et al., 2024b), JRNGC (Zhou et al., 2024) and GCD (Liu et al., 2025). These baselines cover MLP-based, recurrent, convolutional, Jacobian-based, and structure-aware paradigms, and therefore provide a broad comparison for evaluating causal discovery performance on both benchmark and industrial datasets. Forecasting baselines are not included here because most of them do not explicitly produce causal graphs or lag-resolved causal scores, and are therefore not directly comparable under causal discovery metrics.

4.2.3. Experimental details

Since causal discovery in MKJR is performed through forecasting-

based temporal representation learning on causal datasets, the main model settings follow those in Section 4.1.4 unless otherwise stated. For the benchmark causal datasets, we use AUROC and AUPRC as the primary evaluation metrics. AUROC measures the overall ranking quality of predicted causal scores, while AUPRC is particularly informative for sparse and imbalanced causal graphs.

For the industrial process datasets, we use SHD, F1, Recall, and Precision as the evaluation metrics. These metrics assess graph-structure recovery and edge-level detection quality from complementary perspectives. We use a fixed significance-filtering protocol. Specifically, the number of surrogate samples is set to $B_{\text{surr}} = 100$, the significance level is fixed to $\alpha = 0.05$ after Benjamini–Hochberg FDR correction, and the circular-shift offset is sampled with minimum displacement $d_{\min} = L + 1$. These protocol parameters are kept unchanged across all industrial datasets.

4.2.4. Results

We first report the main causal discovery results on four public benchmark datasets, namely VAR, fMRI, DREAM-3, and CausalTime, as summarized in Tables 9–12. Overall, MKJR shows strong and stable performance across synthetic, biomedical, and realistic multivariate causal discovery settings. On the VAR benchmark, MKJR achieves the best overall results under both lag-aggregated and lag-resolved evaluation settings. On the fMRI dataset, MKJR obtains the best AUROC and AUPRC among all compared methods, demonstrating clear advantages in high-dimensional causal analysis. On DREAM-3, MKJR remains the best or among the strongest methods across most subnetworks, indicating good generalization to gene-regulatory causal discovery. On the CausalTime benchmark, MKJR achieves the best results on most metrics, especially on AQI and Medical, while remaining highly competitive on Traffic, where LOCAL slightly outperforms it in AUROC. Compared with Jacobian-based baselines such as JRNGC, MKJR benefits from a stronger nonlinear forecasting backbone and more structured dependence modeling, which together improve the quality and stability of causal discovery.

To further assess practical engineering applicability, we evaluate all compared methods on three industrial process datasets: TE, UF, and DE. The corresponding results are summarized in Tables 13–15. We report SHD, F1, Recall, and Precision, where lower SHD and higher F1/Recall/Precision indicate better causal graph recovery quality. The best and second-best results are highlighted in bold and underline, respectively. Lower SHD and higher F1, Recall, and Precision indicate better causal discovery performance.

On TE, MKJR achieves the best SHD, F1, and Recall, while its Precision is slightly lower than that of LOCAL. On UF, MKJR achieves the best performance across all four metrics. On DE, MKJR again obtains the best SHD and F1, achieves the best Recall jointly with GCD, and remains highly competitive in Precision, although it is lower than that of a few competing methods. These results suggest that MKJR is particularly effective in recovering sparse and practically meaningful causal structures under realistic industrial settings, rather than only on synthetic benchmarks.

4.2.5. Ablation study

To further clarify the contribution of each major component to causal

Table 10
Results of various models on the fMRI dataset in the causal analysis experiment.

Model	AUROC	AUPRC
cMLP	0.875	0.726
cLSTM	0.845	0.644
eSRU	0.743	0.514
JGC	0.912	0.536
JRNGC	0.845	0.707
LOCAL	0.893	0.746
Rhino	0.864	0.653
TCDF	0.796	0.637
GCD	0.915	0.773
MKJR	**0.931**	**0.781**

Table 11
AUROC score results of various models on the DREAM-3 dataset in the causal analysis experiment.

Model	E.Coli 1	E.Coli 2	Yeast 1	Yeast 2	Yeast 3
cMLP	0.644	0.568	0.585	0.506	0.528
cLSTM	0.629	0.609	0.579	0.519	0.555
eSRU	0.660	0.629	0.627	0.575	0.548
JGC	0.504	0.527	0.604	0.553	0.521
JRNGC	0.666	0.678	0.650	0.597	0.560
LOCAL	0.683	0.675	0.602	0.568	0.552
Rhino	0.685	0.680	0.664	0.585	0.567
TCDF	0.614	0.647	0.581	0.556	0.557
GCD	0.676	0.659	0.632	0.593	0.556
MKJR	**0.689**	**0.692**	**0.683**	**0.599**	**0.581**

Table 9
Causal discovery results on the VAR benchmark under lag-aggregated and lag-resolved evaluation settings.

Type	Model	VAR(100,5,10)		VAR(50,5,10)		VAR(10,3,5)	
	Metric	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
Lag-aggregated causal discovery	cMLP	0.973	0.770	0.995	0.893	0.994	0.912
	cLSTM	0.956	0.783	0.964	0.876	0.985	0.874
	eSRU	0.865	0.731	0.847	0.795	0.848	0.796
	JGC	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
	JRNGC	0.997	0.950	**1.000**	0.994	**1.000**	0.998
	LOCAL	0.973	0.930	**1.000**	**1.000**	**1.000**	0.998
	Rhino	0.834	0.628	0.851	0.749	0.795	0.703
	TCDF	0.904	0.753	0.877	0.795	0.848	0.802
	GCD	0.985	0.941	0.992	0.950	0.972	0.965
	MKJR	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
Lag-resolved causal discovery	cMLP	0.940	0.851	0.973	0.929	0.978	0.923
	cLSTM	0.933	0.894	0.952	0.901	0.967	0.915
	eSRU	0.845	0.760	0.896	0.870	0.865	0.913
	JGC	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.998
	JRNGC	0.984	0.960	0.998	0.995	0.996	0.994
	LOCAL	0.965	0.980	0.948	0.986	**1.000**	0.998
	Rhino	0.826	0.692	0.784	0.839	0.865	0.710
	TCDF	0.904	0.753	0.877	0.795	0.848	0.802
	GCD	0.976	0.977	0.995	0.981	0.992	0.978
	MKJR	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**

Table 12
Results of various models on the CausalTime dataset in the causal analysis experiment.

Model	AUROC			AURPC		
	AQI	Traffic	Medical	AQI	Traffic	Medical
cMLP	0.645 ± 0.013	0.633 ± 0.028	0.713 ± 0.009	0.790 ± 0.013	0.585 ± 0.031	0.677 ± 0.012
cLSTM	0.717 ± 0.007	0.603 ± 0.005	0.531 ± 0.015	0.717 ± 0.006	0.358 ± 0.049	0.426 ± 0.025
eSRU	0.823 ± 0.031	0.599 ± 0.019	0.756 ± 0.036	0.722 ± 0.03	0.489 ± 0.031	0.735 ± 0.060
JGC	0.454 ± 0.026	0.423 ± 0.047	0.573 ± 0.028	0.632 ± 0.047	0.280 ± 0.026	0.423 ± 0.045
JRNGC	0.928 ± 0.021	0.729 ± 0.075	0.754 ± 0.017	0.783 ± 0.087	0.594 ± 0.119	0.726 ± 0.134
LOCAL	0.852 ± 0.011	**0.891 ± 0.036**	0.813 ± 0.021	0.670 ± 0.026	0.681 ± 0.035	0.733 ± 0.015
Rhino	0.670 ± 0.036	0.627 ± 0.041	0.652 ± 0.027	0.759 ± 0.033	0.377 ± 0.046	0.490 ± 0.052
TCDF	0.414 ± 0.020	0.503 ± 0.004	0.633 ± 0.038	0.652 ± 0.008	0.363 ± 0.008	0.554 ± 0.031
GCD	0.892 ± 0.021	0.773 ± 0.015	0.919 ± 0.014	0.732 ± 0.034	0.639 ± 0.014	0.904 ± 0.019
MKJR	**0.937 ± 0.008**	0.856 ± 0.008	**0.926 ± 0.007**	**0.894 ± 0.012**	**0.701 ± 0.009**	**0.946 ± 0.011**

Table 13
Industrial causal discovery results on the TE process dataset ($D = 31, T = 1499$).

Model	SHD↓	F1↑	Recall↑	Precision↑
cMLP	37	0.245	0.214	0.286
cLSTM	52	0.161	0.179	0.147
eSRU	63	0.160	0.209	0.128
JGC	47	0.200	0.179	0.227
JRNGC	38	0.303	0.286	0.323
LOCAL	40	0.296	0.250	0.342
Rhino	46	0.221	0.214	0.229
TCDF	42	0.059	0.036	0.167
GCD	35	0.314	0.286	0.348
MKJR	**33**	**0.328**	**0.304**	**0.357**

Table 14
Industrial causal discovery results on the UF process dataset ($D = 17, T = 23132$).

Model	SHD↓	F1↑	Recall↑	Precision↑
cMLP	100	0.254	0.205	0.333
cLSTM	122	0.218	0.205	0.233
eSRU	158	0.325	0.458	0.252
JGC	117	0.270	0.216	0.361
JRNGC	116	0.472	0.640	0.374
LOCAL	121	0.427	0.512	0.366
Rhino	129	0.361	0.434	0.309
TCDF	99	0.075	0.046	0.167
GCD	113	0.511	0.711	0.399
MKJR	**95**	**0.530**	**0.729**	**0.431**

Table 15
Industrial causal discovery results on the DE process dataset ($D = 8, T = 2394$).

Model	SHD↓	F1↑	Recall↑	Precision↑
cMLP	9	0.743	0.650	0.867
cLSTM	10	0.687	0.550	0.917
eSRU	11	0.667	0.550	0.846
JGC	10	0.708	0.650	0.778
JRNGC	7	0.834	0.800	0.870
LOCAL	8	0.815	0.750	0.892
Rhino	10	0.741	0.700	0.786
TCDF	16	0.333	0.200	**1.000**
GCD	6	0.850	**0.850**	0.850
MKJR	**5**	**0.876**	**0.850**	1.943

discovery, we perform a unified one-factor-at-a-time ablation study on VAR100, fMRI, and Lorenz-96, as summarized in Table 16. Specifically, we remove the KAN module, the Mamba backbone, the local gate G_{loc}, the global gate G_{glo}, the sensitivity regularizer R_{sens}, the structured sparsity term R_{str}, and the whole Jacobian-based objective, respectively. Here, w/o Jacobian denotes removing both R_{sens} and R_{str} simultaneously.

Several observations can be made. First, removing KAN or Mamba consistently degrades performance, confirming that nonlinear variable-wise representation and long-range temporal modeling both contribute to causal discovery quality. Their impact is particularly visible on fMRI, where high-dimensional temporal dependence is more difficult to model. Second, removing the gating components causes only moderate degradation, which suggests that they mainly play a supportive role in refining feature conditioning rather than dominating causal inference by themselves. Third, removing R_{sens}, or especially R_{str} leads to much larger drops, indicating that target-specific routing and Jacobian-related regularization are more directly responsible for the causal-analysis capability of MKJR. Finally, removing the whole Jacobian objective results in the largest overall degradation across all three datasets, showing that the Jacobian-based training objective is a key factor for robust and discriminative causal discovery.

Overall, the ablation results suggest that forecasting-oriented modules and causality-oriented modules play different but complementary roles: KAN and Mamba provide the representation foundation, while the routing and Jacobian-related components contribute more directly to causal discrimination.

4.2.6. Hyperparameter sensitivity analysis

To examine the robustness of the causal regularization, we further analyze the sensitivity of three key hyperparameters, namely λ_{str}, λ_{sens}, and n_{proj}, on the fMRI and Lorenz-96 datasets. For brevity, we report AUROC in Table 17, since it provides the clearest and most stable comparison across settings.

As shown in Table 17, all three hyperparameters exhibit a non-monotonic but interpretable trend, with the best results generally achieved at moderate values. For both λ_{str} and λ_{sens}, performance improves as the coefficient increases from very small values to a moderate level, indicating that appropriate Jacobian-related regularization helps the model recover cleaner and more discriminative causal structure. However, excessively large coefficients tend to over-constrain the learned dependence patterns and lead to performance degradation. A similar trend is also observed for n_{proj}, since very small values lead to unstable stochastic Jacobian estimates, whereas overly large values yield only marginal improvement. Accordingly, the best overall performance in our setting is obtained with $n_{proj} = 4$.

4.2.7. Complexity and efficiency analysis

To assess model efficiency, we first summarize the dominant training complexity of the compared causal discovery models in Table 18. Here, e denotes the number of training epochs, M the number of variables, L the input window length, and N_1 the hidden width. For conciseness, method-specific constants in baseline models are absorbed into constant factors.

As shown in Table 18, compact baselines such as cMLP and GCD enjoy lower theoretical complexity because they rely on relatively simple backbones. By contrast, models based on recurrent temporal modeling, convolutional lag modeling, or explicit structural estimation usually involve heavier terms such as MLN_1^2 or M^2L. MKJR is more expensive than the lightest baselines because it combines a KAN-based

Table 16
Unified ablation study of MKJR for causal discovery on VAR100, fMRI, and Lorenz-96.

Variants	VAR100 (Avg.)		fMRI (Avg.)		Lorenz-96 (Avg.)	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
w/o KAN	0.991 ± 0.004	0.923 ± 0.006	0.860 ± 0.014	0.815 ± 0.005	0.942 ± 0.016	0.884 ± 0.002
w/o Mamba	0.989 ± 0.002	0.917 ± 0.014	0.852 ± 0.011	0.806 ± 0.006	0.936 ± 0.056	0.872 ± 0.016
w/o G^{loc}	0.992 ± 0.004	0.928 ± 0.015	0.861 ± 0.047	0.817 ± 0.007	0.939 ± 0.011	0.878 ± 0.004
w/o G^{glo}	0.991 ± 0.014	0.925 ± 0.013	0.858 ± 0.021	0.812 ± 0.008	0.941 ± 0.016	0.882 ± 0.020
w/o R_{sens}	0.987 ± 0.032	0.910 ± 0.020	0.845 ± 0.006	0.776 ± 0.011	0.929 ± 0.036	0.857 ± 0.056
w/o R_{str}	0.894 ± 0.028	0.696 ± 0.068	0.815 ± 0.030	0.735 ± 0.042	0.891 ± 0.024	0.836 ± 0.016
w/o Jacobian	0.871 ± 0.041	0.654 ± 0.030	0.785 ± 0.029	0.692 ± 0.079	0.862 ± 0.020	0.741 ± 0.041
MKJR	**1.000** ± 0.000	**1.000** ± 0.000	**0.942** ± 0.005	**0.831** ± 0.008	**0.999 ± 0.001**	**0.999 ± 0.001**

Table 17
Sensitivity of MKJR to λ_{str}, λ_{sens}, and n_{proj} on fMRI and Lorenz-96 (AUROC).

λ_{str}	0.01	0.02	0.05	0.10	0.20	0.50
	AUROC	AUROC	AUROC	AUROC	AUROC	AUROC
fMRI	0.812 ± 0.008	0.854 ± 0.014	0.876 ± 0.004	**0.942 ± 0.030**	0.913 ± 0.030	0.873 ± 0.004
Lorenz-96	0.815 ± 0.021	0.867 ± 0.041	0.979 ± 0.032	**0.999 ± 0.001**	0.959 ± 0.002	0.912 ± 0.013
λ_{sens}	0.01	0.02	0.05	0.10	0.20	0.50
fMRI	0.842 ± 0.030	0.867 ± 0.016	0.936 ± 0.011	**0.942 ± 0.030**	0.919 ± 0.030	0.873 ± 0.006
Lorenz-96	0.812 ± 0.008	0.846 ± 0.011	0.958 ± 0.015	**0.999 ± 0.001**	0.976 ± 0.016	0.936 ± 0.056
n_{proj}	1	2	3	4	5	6
fMRI	0.815 ± 0.005	0.845 ± 0.006	0.902 ± 0.021	**0.942 ± 0.030**	0.931 ± 0.007	0.917 ± 0.028
Lorenz-96	0.907 ± 0.014	0.922 ± 0.011	0.927 ± 0.030	**0.999 ± 0.001**	0.936 ± 0.047	0.924 ± 0.016

Table 18
Dominant training complexity of the compared causal discovery models.

Model	Time Complexity
cMLP	$\mathcal{O}(eMN_1^2)$
cLSTM	$\mathcal{O}(eM(N_1^2 + MN_1))$
eSRU	$\mathcal{O}(eM(N_1^2 + MN_1))$
JGC	$\mathcal{O}(eMLN_1^2) + O(M^2L)$
JRNGC	$\mathcal{O}(eMN_1^2) + O(M^2L)$
LOCAL	$O(eM^2L)$
Rhino	$\mathcal{O}(e(M^2L + MN_1^2))$
TCDF	$\mathcal{O}(eMLN_1^2)$
GCD	$\mathcal{O}(eN_1^2 + M^2)$
MKJR	$\mathcal{O}(e(LN_1 + MN_1^2))$

encoder-decoder with a shared dual-stream Mamba backbone and further introduces Jacobian regularization during training. However, MKJR still does not fall into the worst complexity regime.

To complement the asymptotic analysis, Table 19 further reports the parameter size and practical training cost of the compared models on the TE, UF, and DE industrial datasets. These empirical results help verify whether the theoretical complexity trends are consistent with actual runtime behavior.

Overall, MKJR is not the lightest model, but it is also not the most computationally expensive one in practice. On TE and UF, its training cost remains below several heavier recurrent or structure-aware baselines, while on DE it becomes relatively heavier but still does not represent the worst case. These results indicate that MKJR achieves a reasonable balance between computational overhead and modeling capability.

Table 19
Parameter size and practical training cost of the compared causal discovery models on the TE, UF, and DE industrial datasets.

Datasets	Model	Total parameters	Training time per epoch (s)	Total training time (s)
TE process ($M = 31$)	cMLP	486715	0.057	114.44
	cLSTM	1652331	0.142	285.22
	eSRU	218302	0.133	267.56
	JGC	258013	0.106	213.484
	JRNGC	97429	0.072	144.43
	LOCAL	79473	0.065	130.59
	Rhino	129734	0.179	359.25
	TCDF	70184	0.105	210.39
	GCD	64543	0.036	72.22
	MKJR	562074	0.127	255.17
UF process ($M = 17$)	cMLP	73954	0.042	85.02
	cLSTM	810916	0.809	1618.82
	eSRU	55096	0.180	360.59
	JGC	59273	0.139	278.56
	JRNGC	74862	0.092	184.83
	LOCAL	60481	0.062	124.43
	Rhino	94875	0.144	289.15
	TCDF	46052	0.055	110.28
	GCD	35857	0.034	68.26
	MKJR	397481	0.103	206.72
DE process ($M = 8$)	cMLP	67208	0.022	43.95
	cLSTM	352808	0.041	82.64
	eSRU	24496	0.075	151.28
	JGC	25736	0.058	116.97
	JRNGC	30495	0.049	98.83
	LOCAL	27913	0.045	90.86
	Rhino	62701	0.029	58.31
	TCDF	18112	0.095	191.67
	GCD	17416	0.017	35.58
	MKJR	210524	0.093	187.51

4.3. Discussion

The results suggest that the effectiveness of MKJR comes from the interaction of three components: KAN-based nonlinear representation, dual-stream Mamba sequence modeling, and dependence-aware Jacobian regularization. Together, they allow the framework to capture variable-wise temporal structure and cross-variable lagged dependencies within one forecasting model.

At the same time, MKJR does not dominate every dataset or metric, which indicates that specialized methods may still be advantageous in certain settings. Its main value therefore lies not in replacing all existing approaches, but in providing a unified and practically effective framework that supports both accurate long-horizon forecasting and graph-oriented variable-level diagnostics.

5. Conclusion and limitations

Conclusion. This work proposes MKJR, a unified framework for jointly supporting long-term time series forecasting and Granger-style

causal analysis. Rather than simply placing prediction and analysis within the same model, MKJR organizes predictive modeling, dependence-aware regularization, and lag-resolved dependence analysis around a single multivariate predictor. Specifically, the framework combines a KAN-based encoder-decoder, an asymmetric bidirectional dual-stream Mamba backbone, and a random-projection Jacobian regularization mechanism that does not explicitly construct a dense Jacobian during training. This design enables the model to learn variable-wise nonlinear temporal structure and long-range cross-variable dependencies, while also providing informative internal signals for subsequent lag-resolved dependence analysis.

The main contribution of this study lies in establishing a predictor-centered unified design in which long-horizon forecasting and lag-resolved Granger-style dependence analysis are no longer handled by separate models or loosely connected procedures, but are instead built on the same trained predictor and its learned representations. Within this framework, the structured representations, readout preferences, and sensitivity information learned by the predictor jointly support dependence analysis. At the same time, Jacobian-based regularization encourages smoother and more target-relevant lagged predictive dependencies during training, which improves forecasting robustness and enhances the reliability of subsequent dependence analysis. In this sense, MKJR does not merely combine two functionalities, but provides a coordinated design centered on the same predictor.

Experimental results across multiple forecasting and causal-analysis datasets show that MKJR achieves strong overall performance in long-term forecasting and remains competitive in Granger-style dependence analysis and graph recovery tasks. These results indicate that the proposed framework not only improves long-horizon predictive quality, but also provides informative variable-level lagged dependence evidence that can support graph-oriented diagnosis and mechanism analysis in complex multivariate systems. In addition, the experiments, ablation studies, and sensitivity analyses suggest that the KAN-Mamba architecture and the Jacobian-based mechanism do not act independently, but interact in a complementary manner across representation learning, dependence regularization, and relation recovery. This empirical pattern also supports the central idea of mutual reinforcement in MKJR.

Limitations and future works. First, the richer structural and sensitivity information used by MKJR comes with a moderate increase in computation during training and post-training analysis, especially in high-dimensional settings or under long lag ranges. Second, the recovered relations are better interpreted as Granger-style predictive dependence derived from post-training analysis rather than as formal structural causality. Accordingly, the final graph may retain some dependence on the choice of scoring and significance-filtering strategy. Finally, although MKJR has been evaluated on industrial datasets, broader validation on larger and more diverse real-world systems would further clarify its practical scope.

Future work will focus on addressing these limitations. In particular, we plan to improve the computational efficiency of Jacobian-based regularization and sensitivity analysis, enhance robustness under abrupt local fluctuations, strong noise, and severe distribution shifts, and develop more end-to-end and structure-aware relation recovery strategies. We also plan to incorporate richer structural priors when explicit topology or evolving interaction structures are available, so as to further improve the practical value of MKJR in complex engineering systems.

CRediT authorship contribution statement

Yanan Zhang: Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Bo Feng:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation. **Zan Chen:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Shen Zhang:** Writing – review & editing, Validation, Conceptualization.

Availability of data and material

The sources of public datasets involved in the experiments are as follows.

(1) ETT-small: https://github.com/zhouhaoyi/ETDataset/
(2) Traffic: https://pems.dot.ca.gov/
(3) Electricity: https://archive.ics.uci.edu/dataset/321/electricityloaddiagrams20112014/
(4) Weather: https://www.bgc-jena.mpg.de/wetter/
(5) VAR: https://github.com/prakashtewari/Python-VAR/
(6) fMRI: https://openfmri.org/dataset/
(7) Lorenz96: https://docs.dart.ucar.edu/en/latest/models/lorenz_96/readme.html/
(8) CausalTime: http://www.causaltime.cc/
(9) Dream3: https://www.synapse.org/Synapse:syn3033083/wiki/74370/
(10) TE: https://github.com/giovanniMen/CPCaD-Bench/tree/main/TenneseeEastman.
(11) UF: https://github.com/giovanniMen/CPCaD-Bench/tree/main/UltraProcessed_Food
(12) DE: https://home.isr.uc.pt/~fasouza/datasets.html.

Declaration of generative AI and AI-assisted technologies

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to improve the language clarity and readability of the text (e.g., grammar, spelling, and phrasing). After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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